



US 20250281787A1

(19) **United States**

(12) **Patent Application Publication**
BANERJI

(10) **Pub. No.: US 2025/0281787 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **TOUCH SEQUENCE MAT REHABILITATION
SYSTEM**

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(21) Appl. No.: **19/073,022**

(22) Filed: **Mar. 7, 2025**

Related U.S. Application Data

(60) Provisional application No. 63/562,270, filed on Mar.
7, 2024.

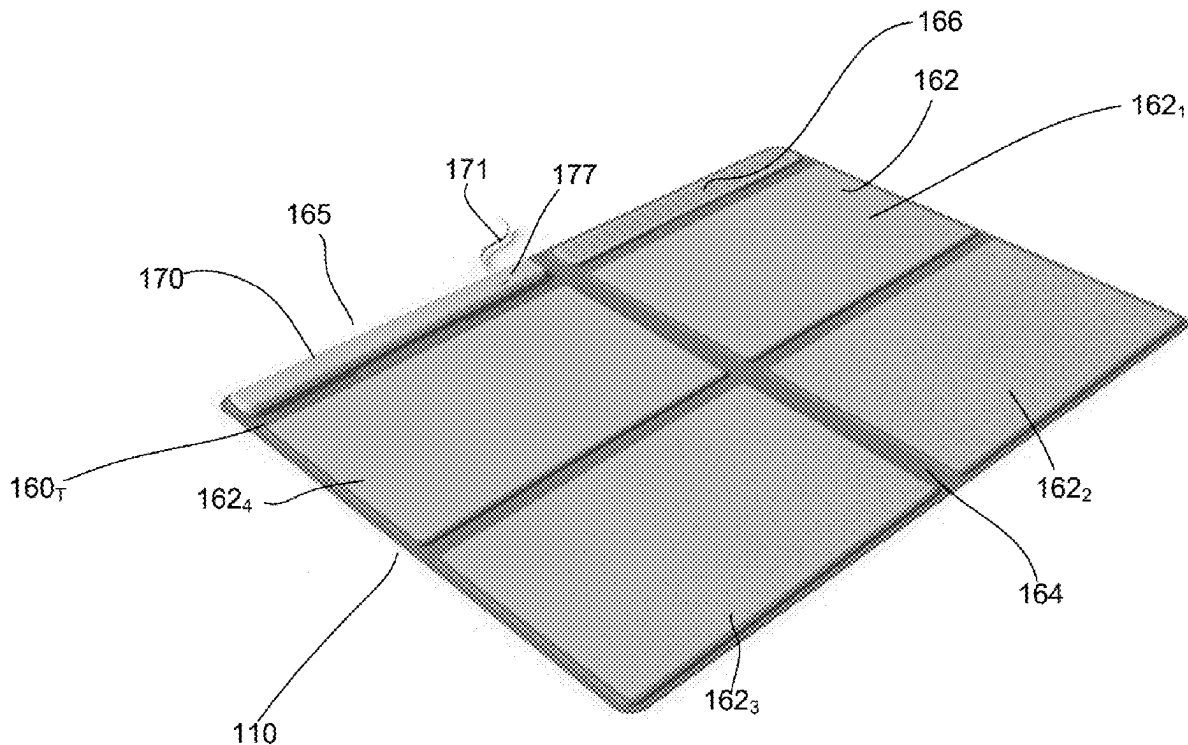
Publication Classification

(51) **Int. Cl.**
A63B 6/00 (2006.01)

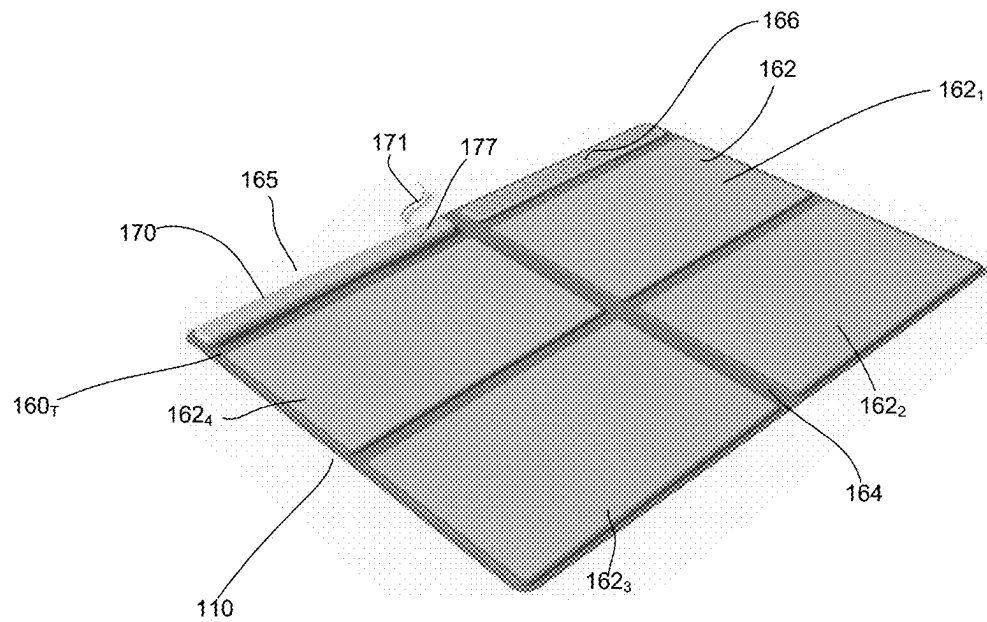
(52) **U.S. Cl.**
CPC **A63B 6/00** (2013.01); **A63B 2220/56**
(2013.01); **A63B 2225/15** (2013.01)

(57) **ABSTRACT**

A flexible touch sequence mat for rehabilitating patients having mobility, functional, coordination and cognitive issues stemming from neurological damage or disorder is disclosed. The touch sequence mat includes a function region with pressure sensors collecting spatial, temporal and magnitude pressure information when a patient is performing prescribed exercises.



100



100
Fig. 1a

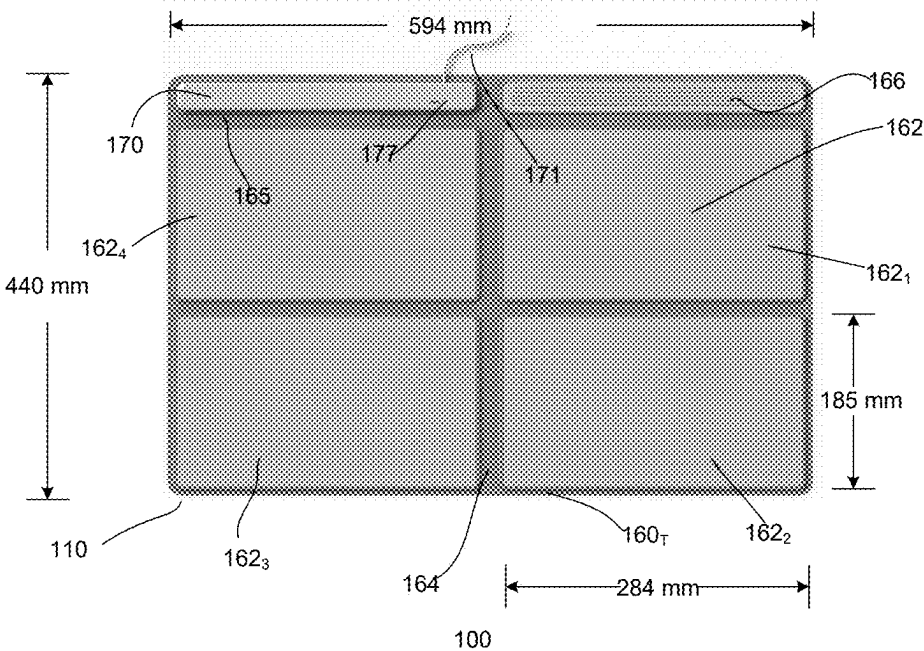


Fig. 1b

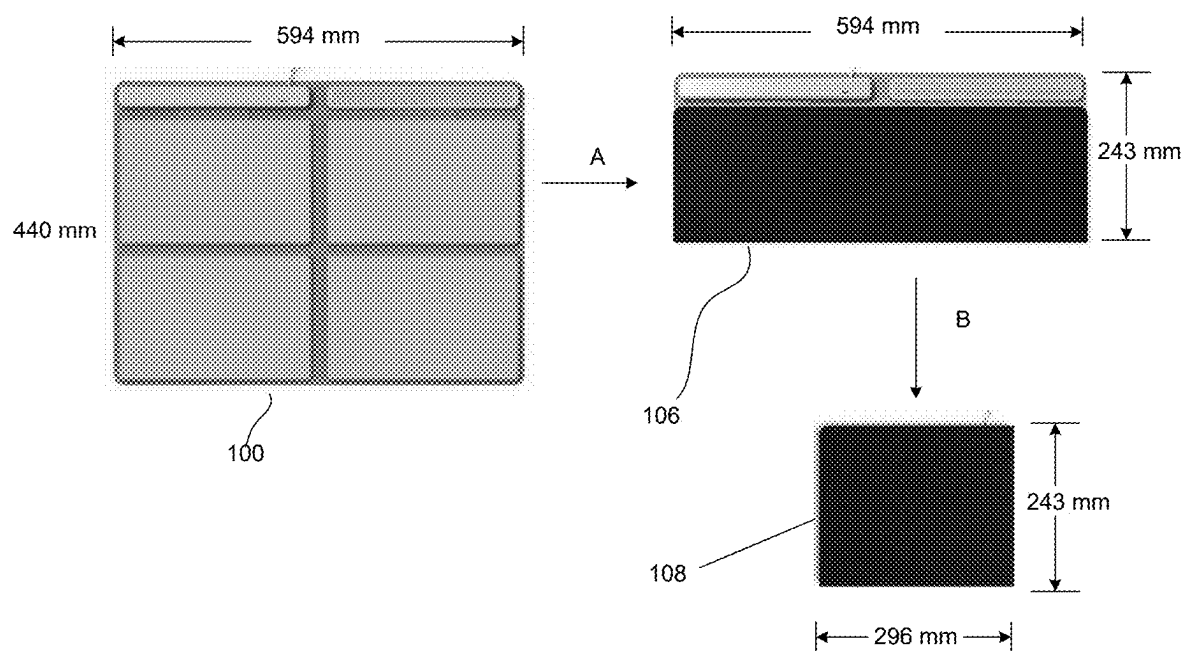
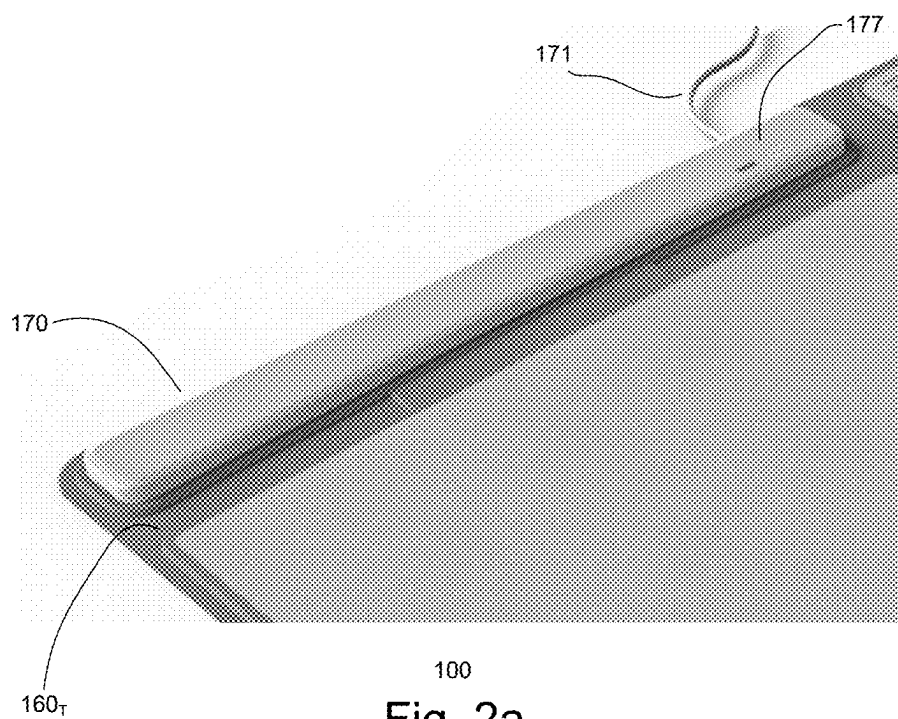


Fig. 1d



100
Fig. 2a

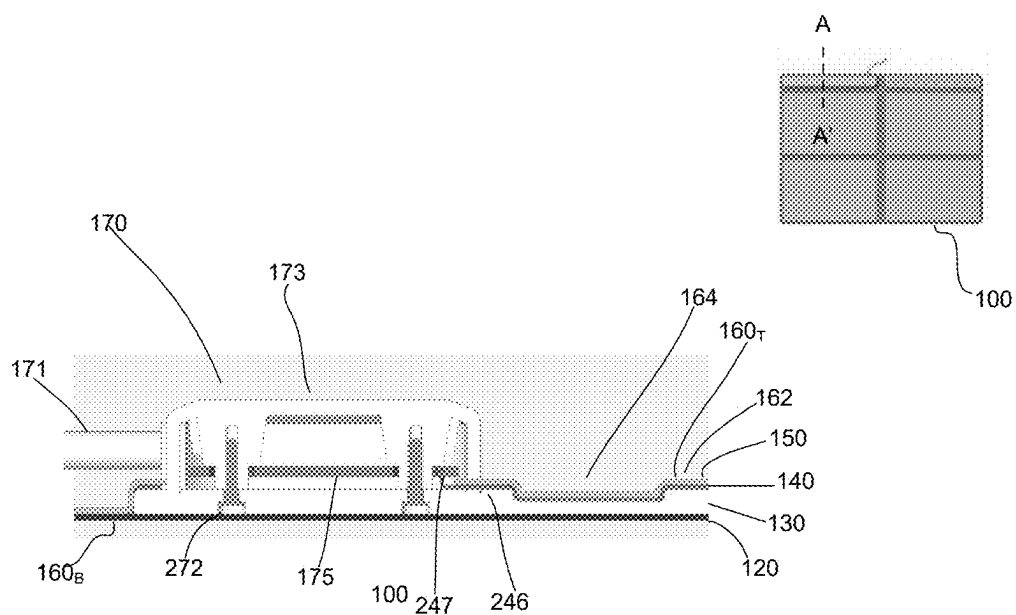


Fig. 2b

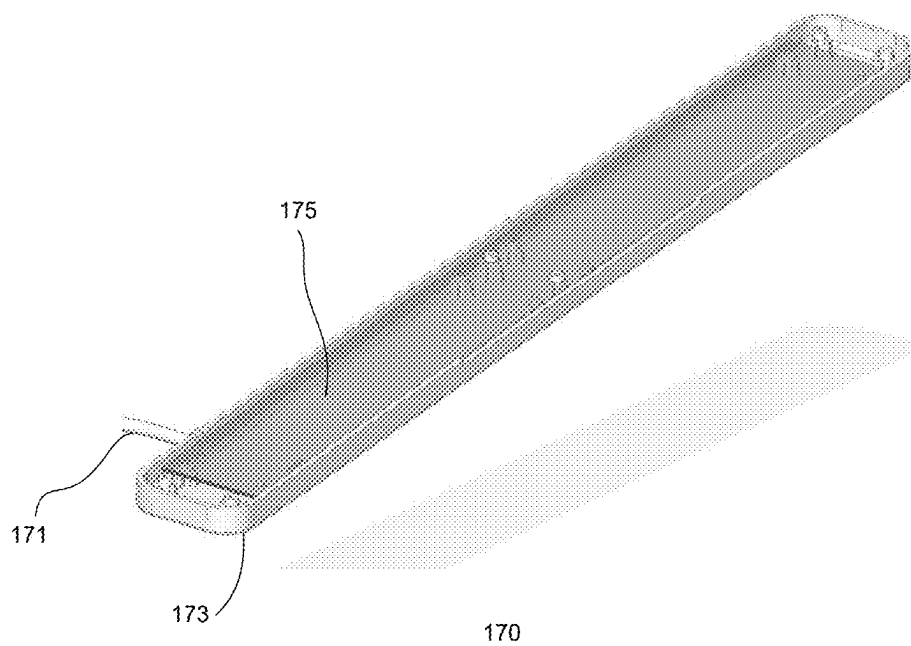
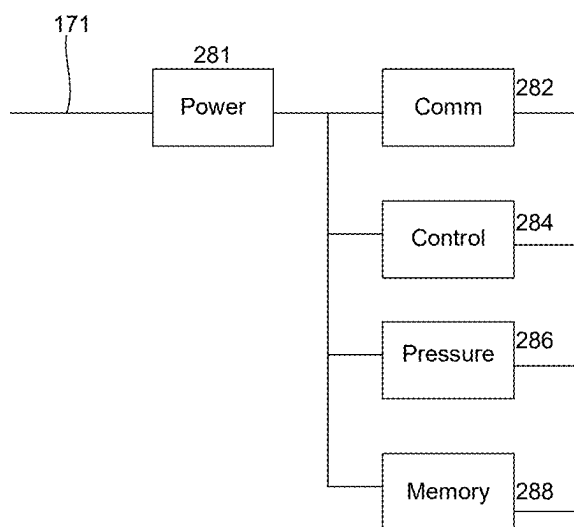


Fig. 2c



175
Fig. 2d

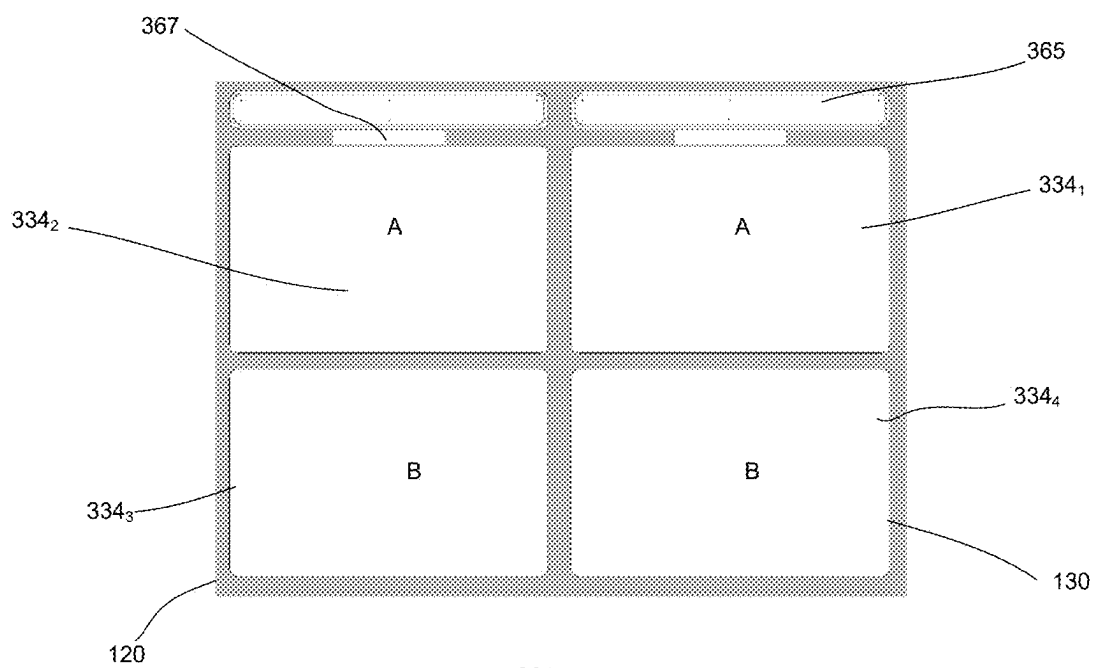
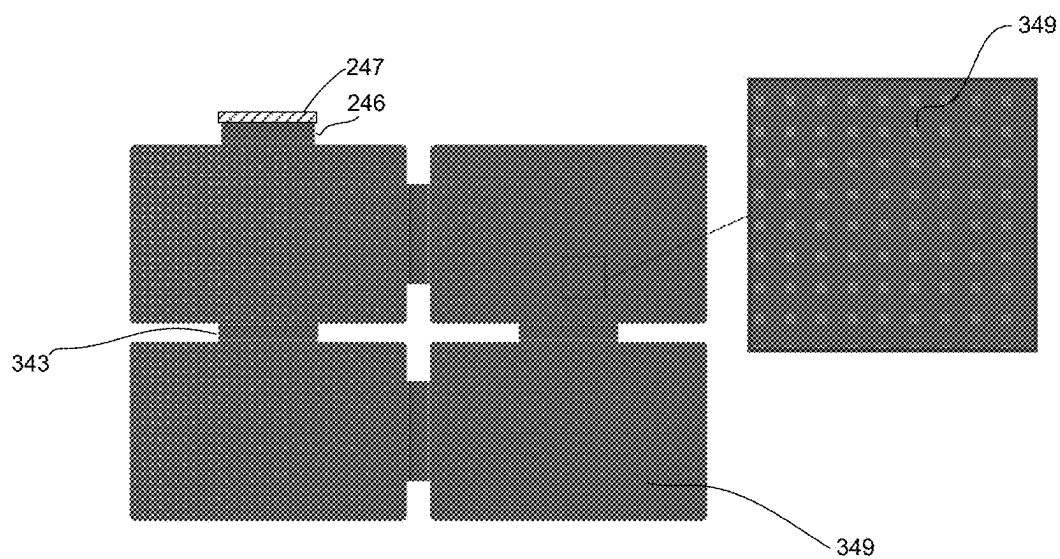
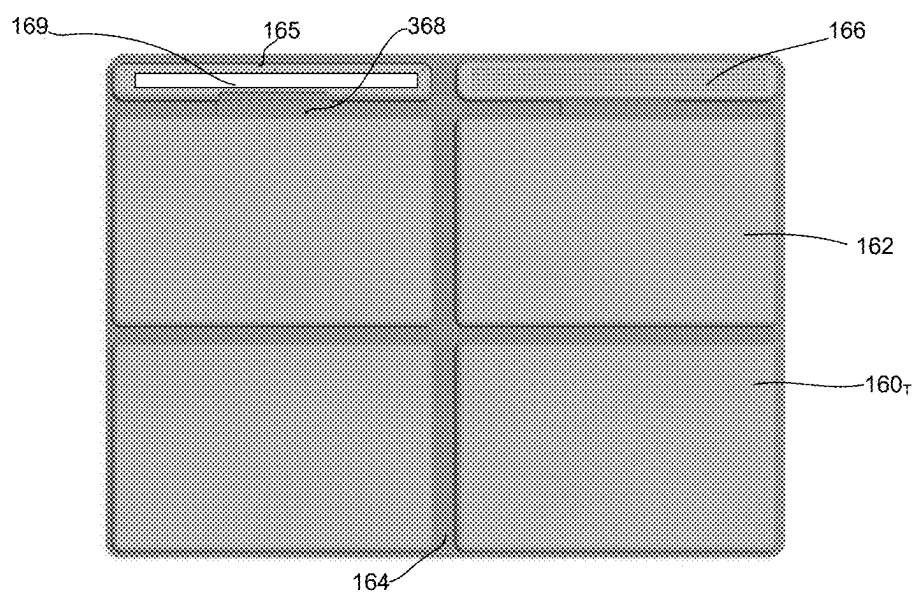


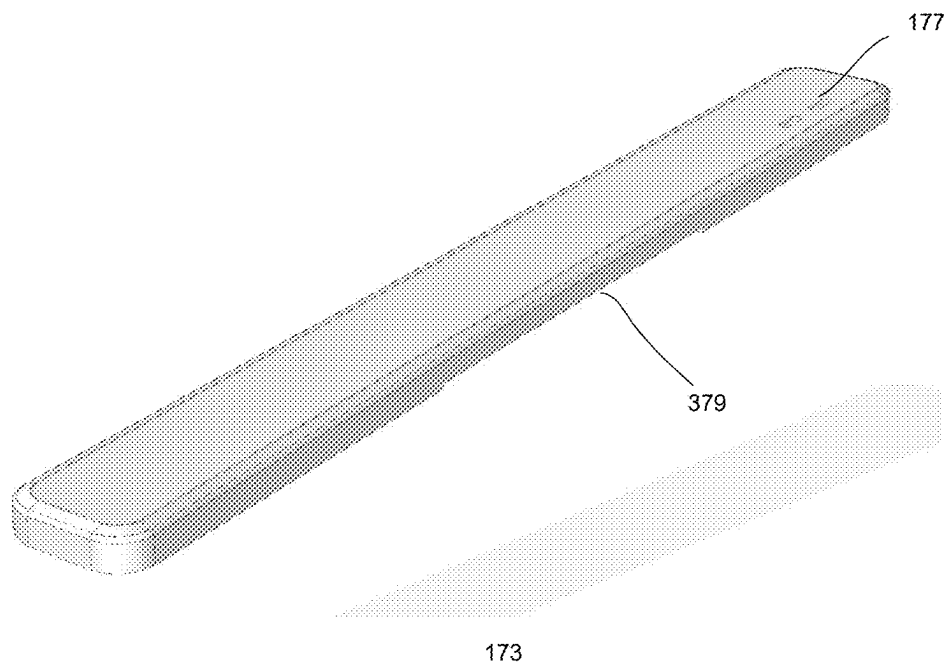
Fig. 3a



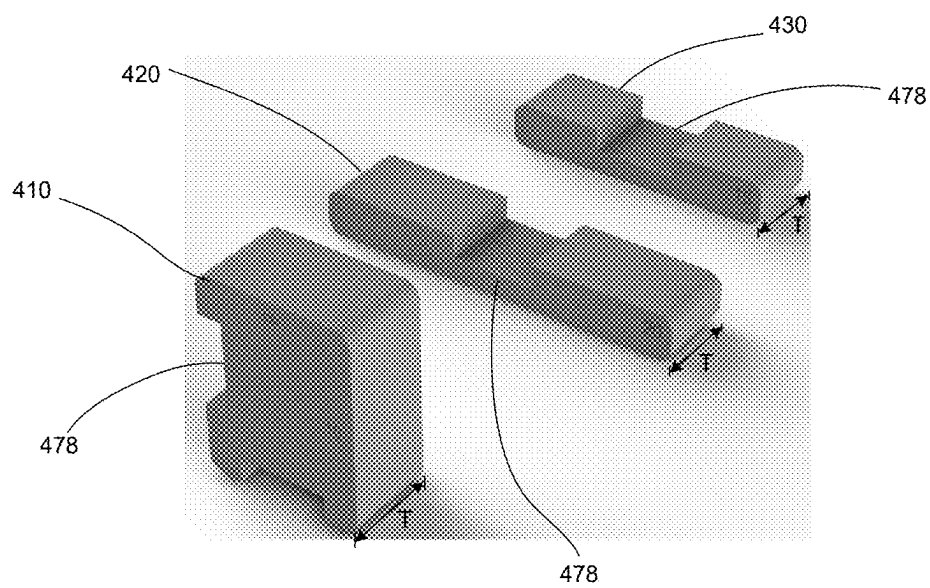
140
Fig. 3b



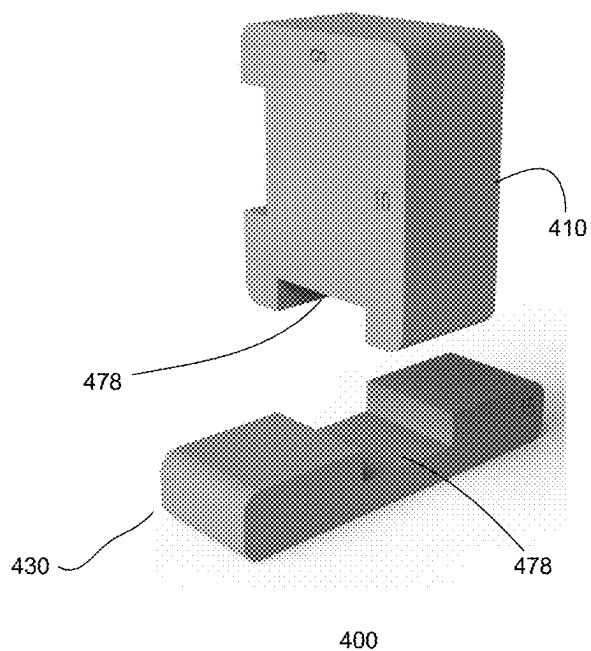
150
Fig. 3c



173
Fig. 3d



400
Fig. 4a



400
Fig. 4b

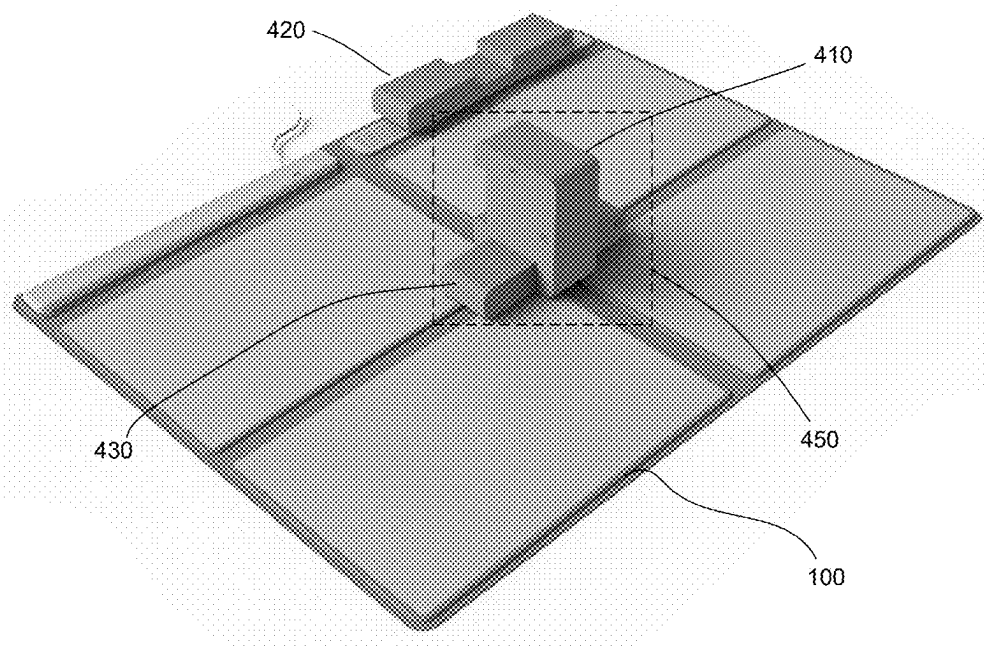
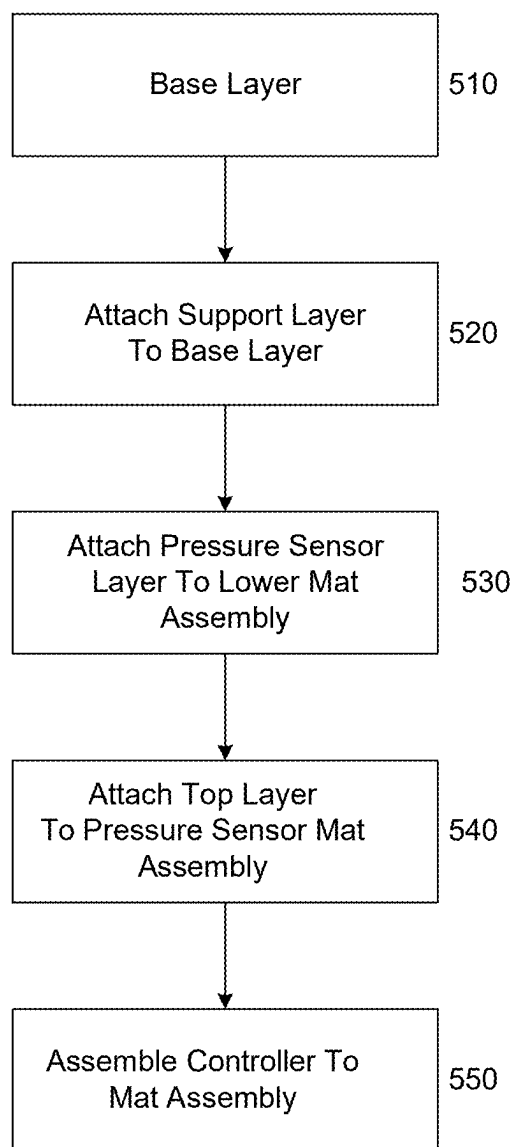
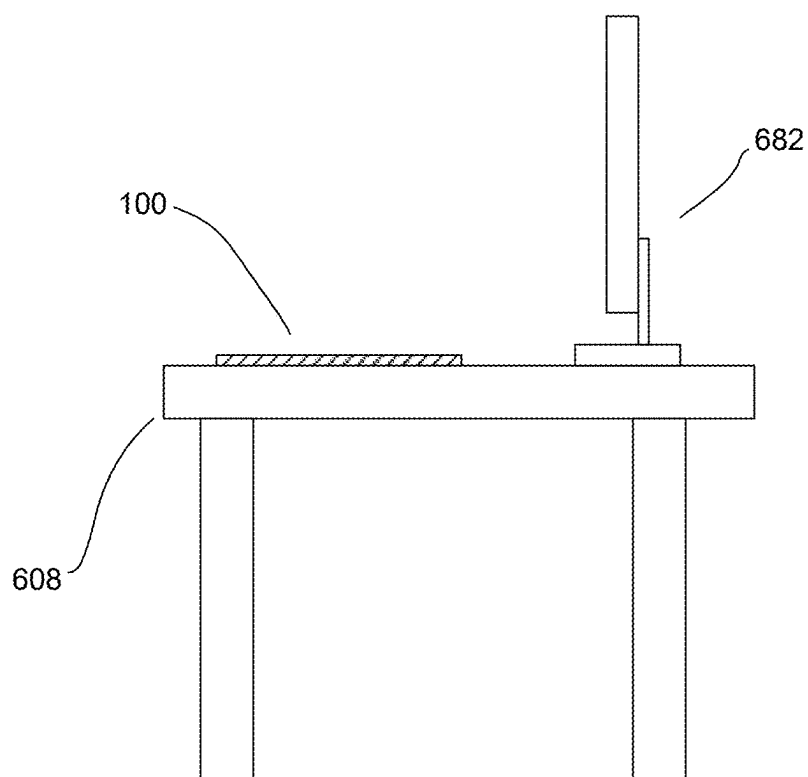


Fig. 4c



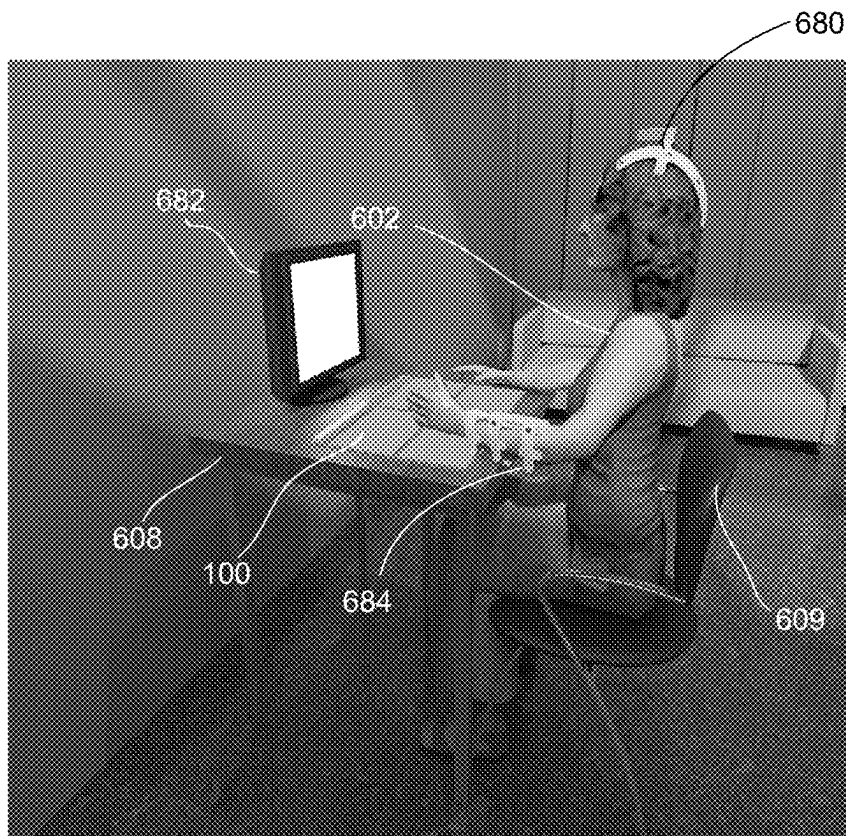
500

Fig. 5

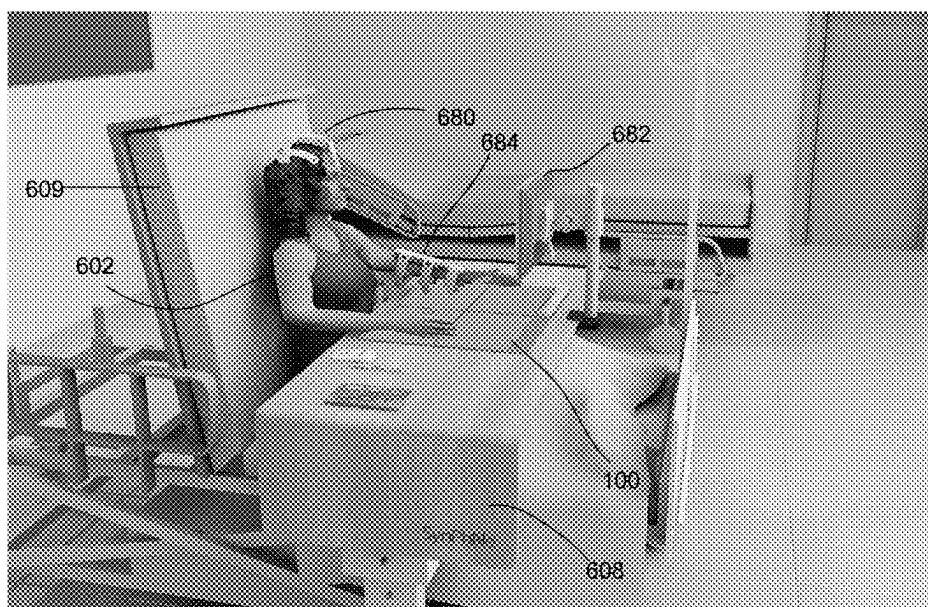


600

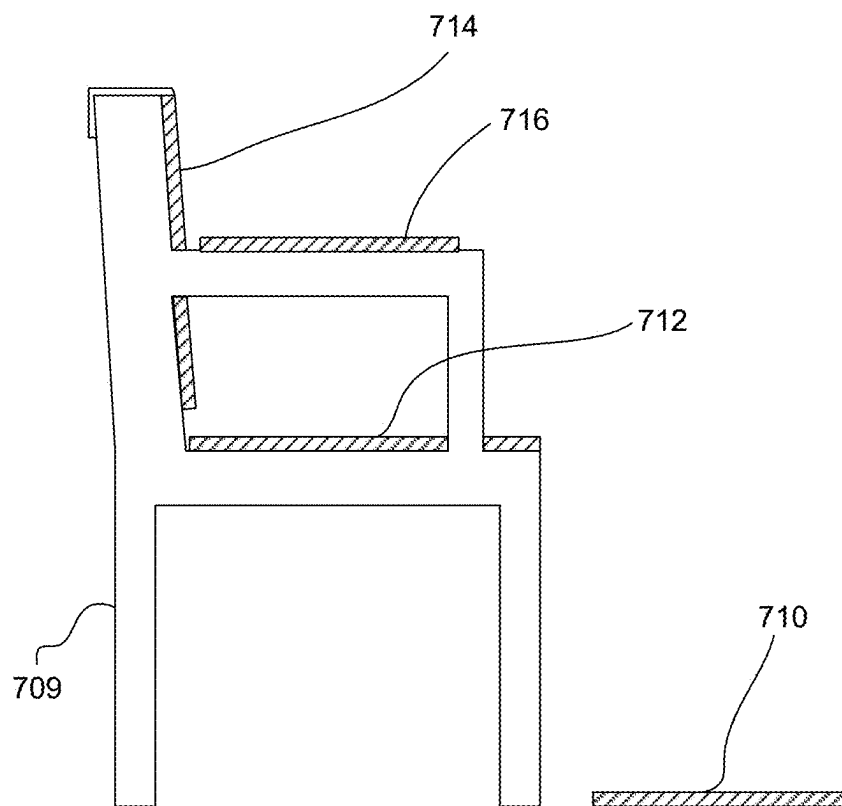
Fig. 6a



600
Fig. 6b

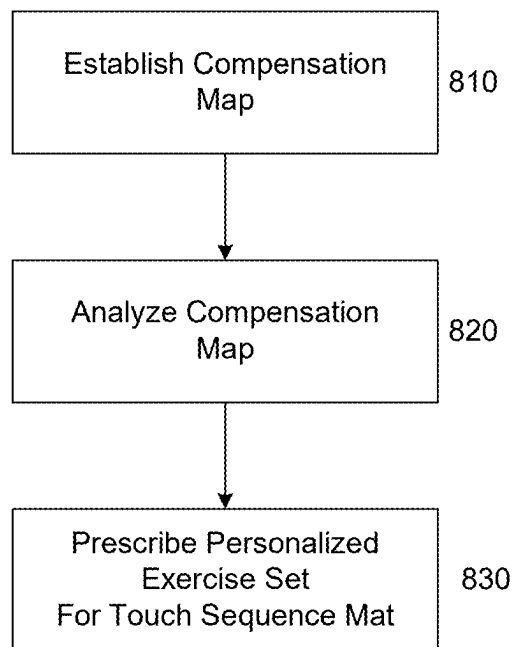


600
Fig. 6c



700

Fig. 7



800

Fig. 8

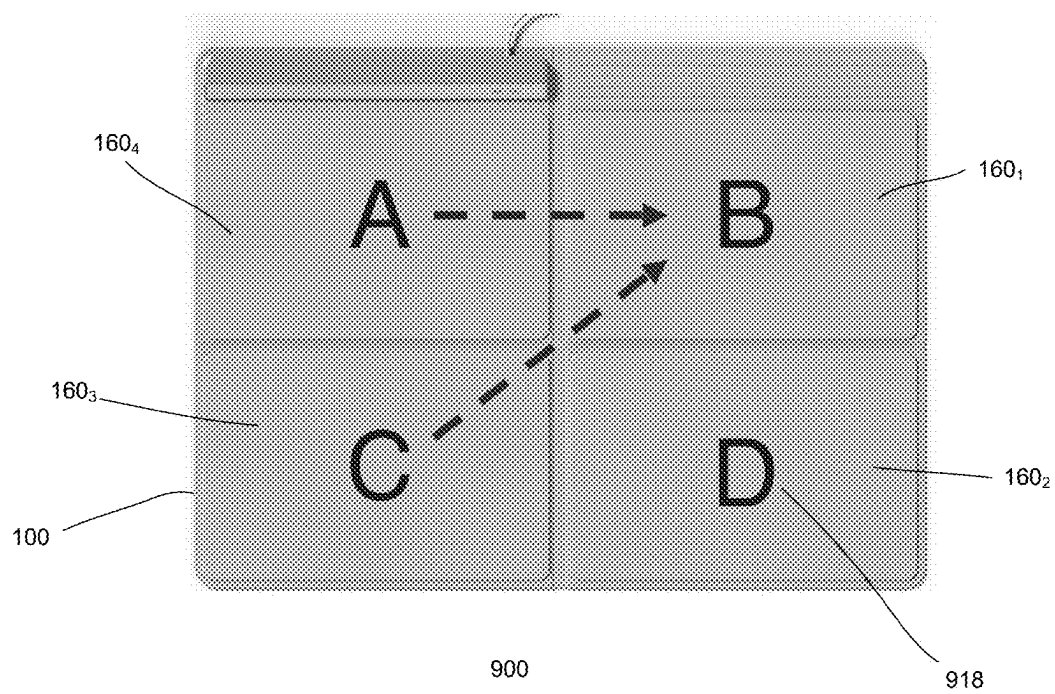
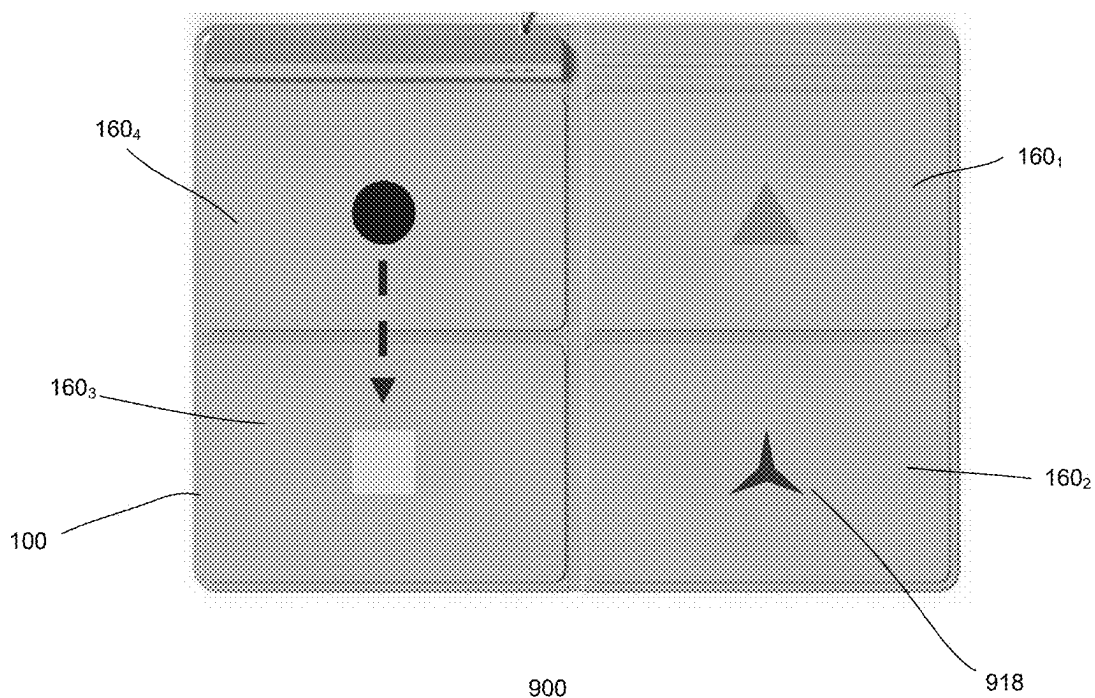
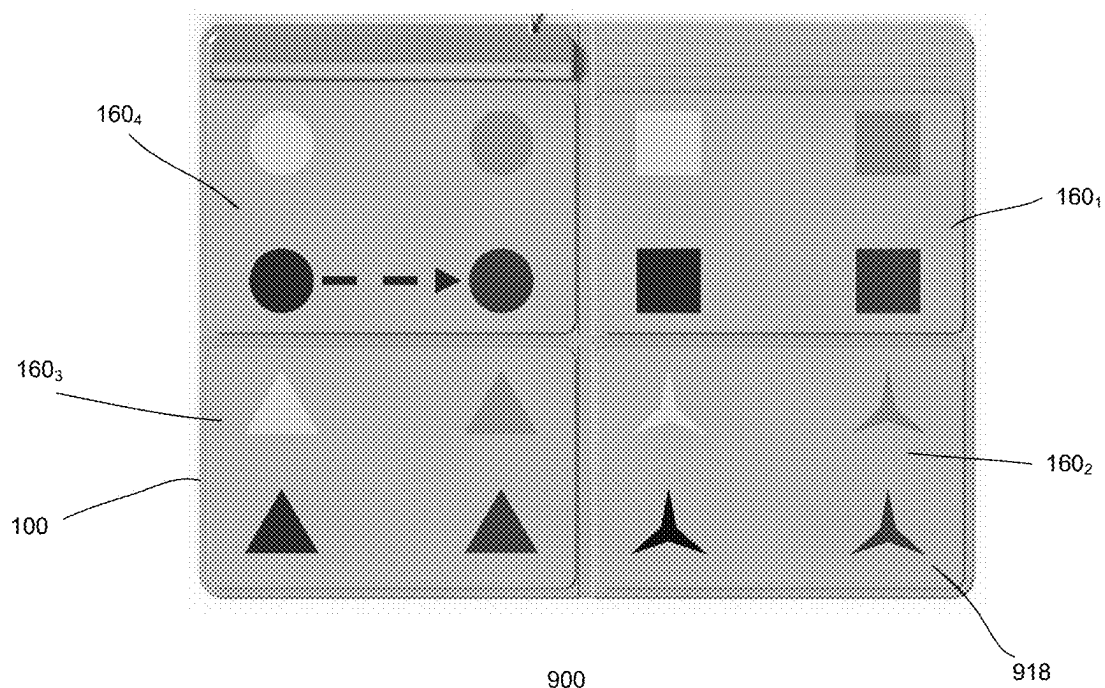


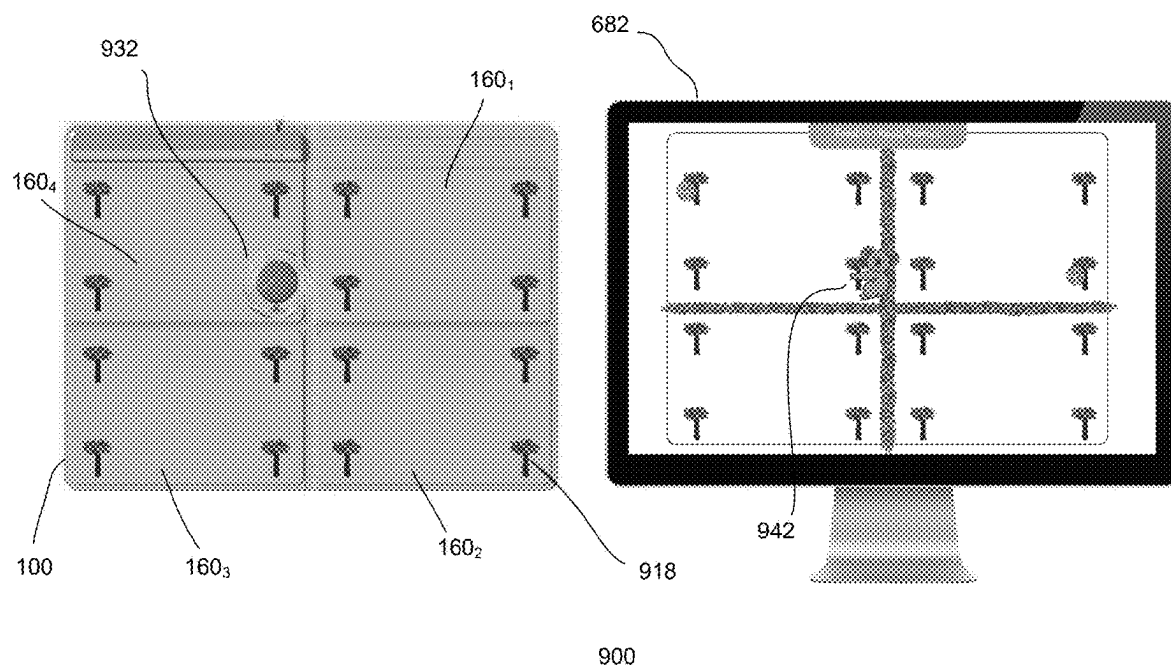
Fig. 9a



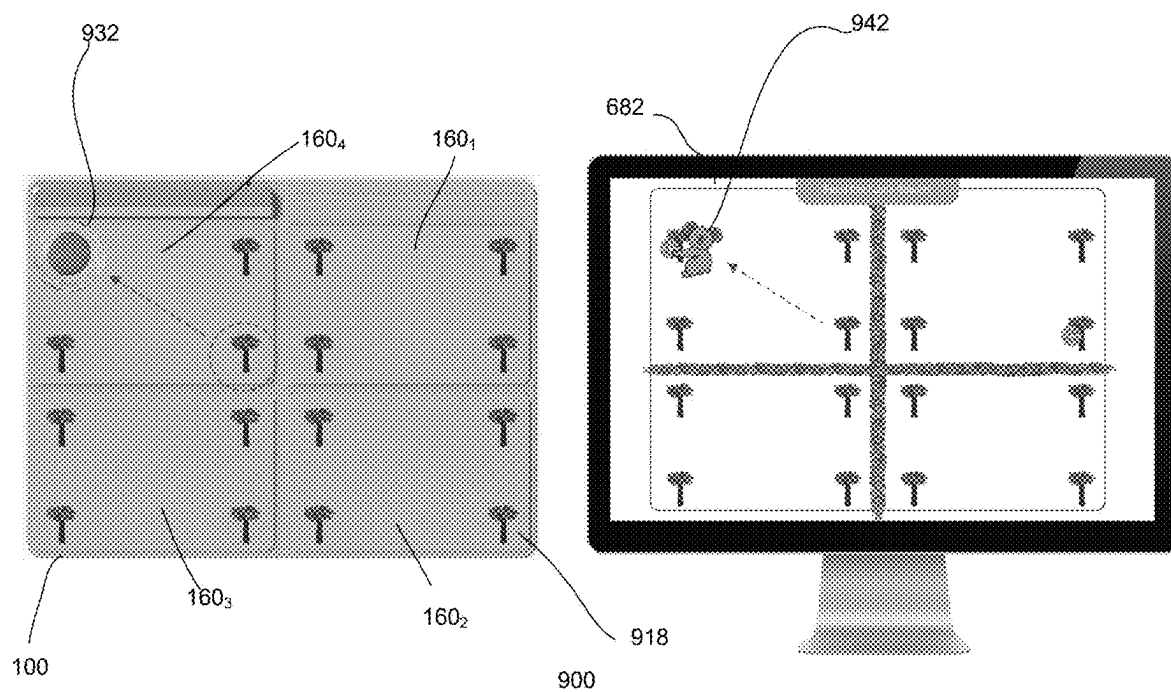
900
Fig. 9b

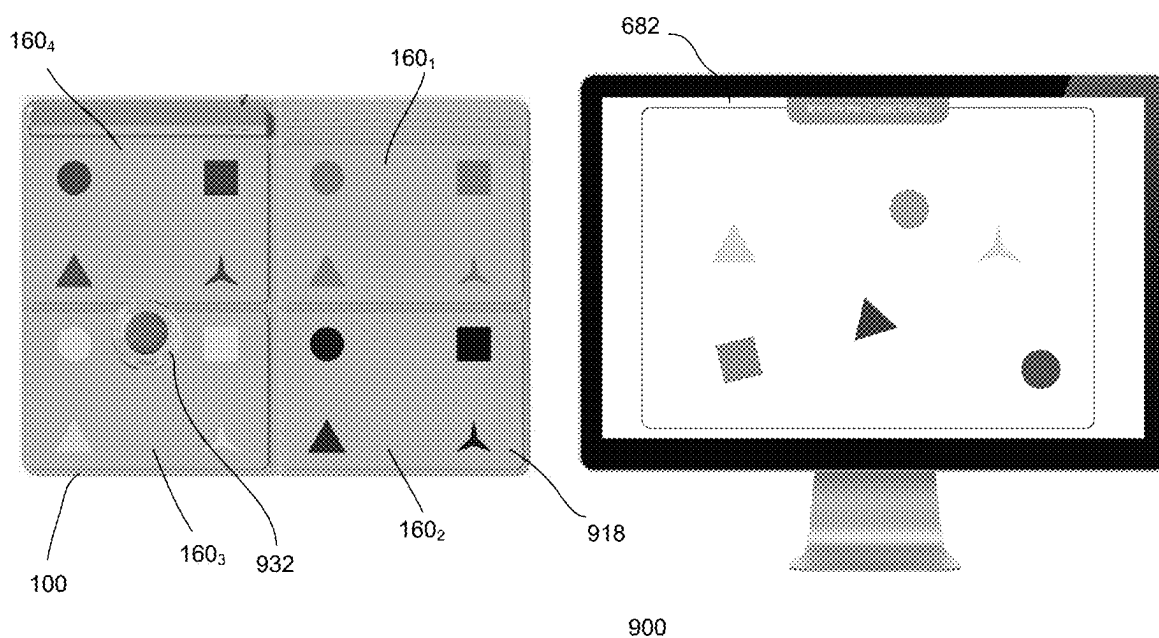


900
Fig. 9c



900
Fig. 9d1





900
Fig. 9e1

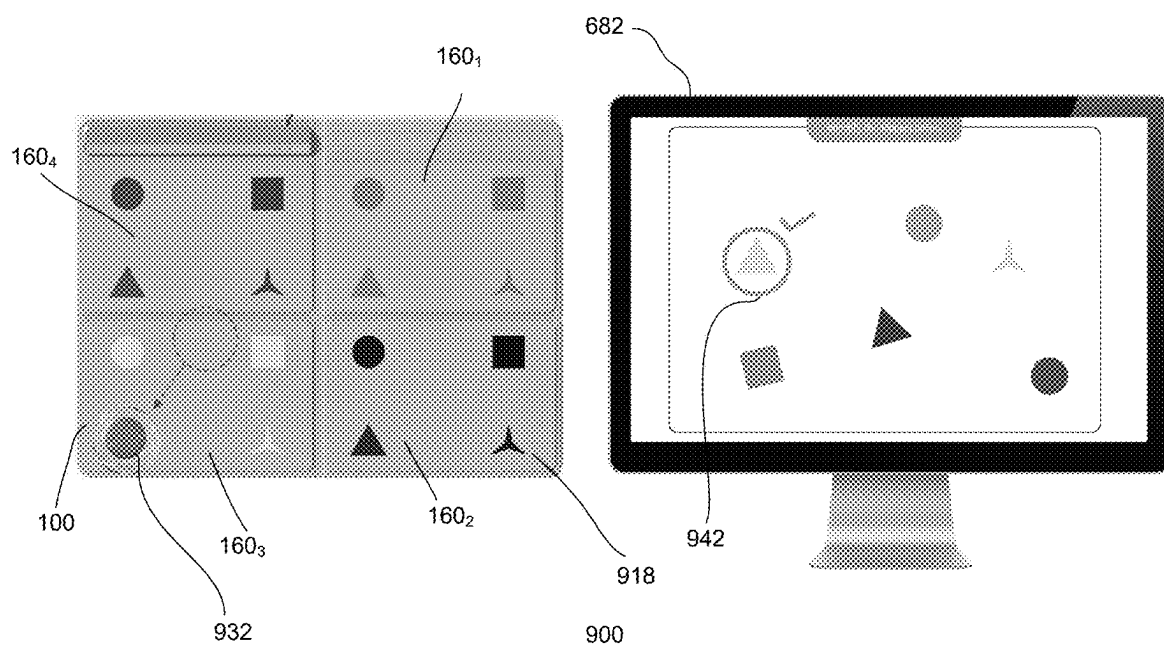
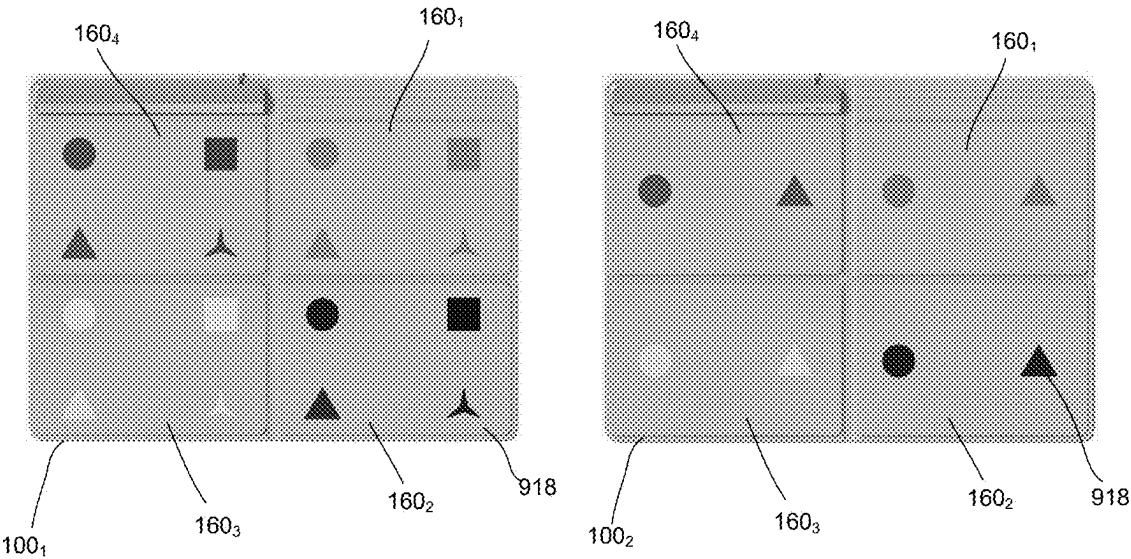
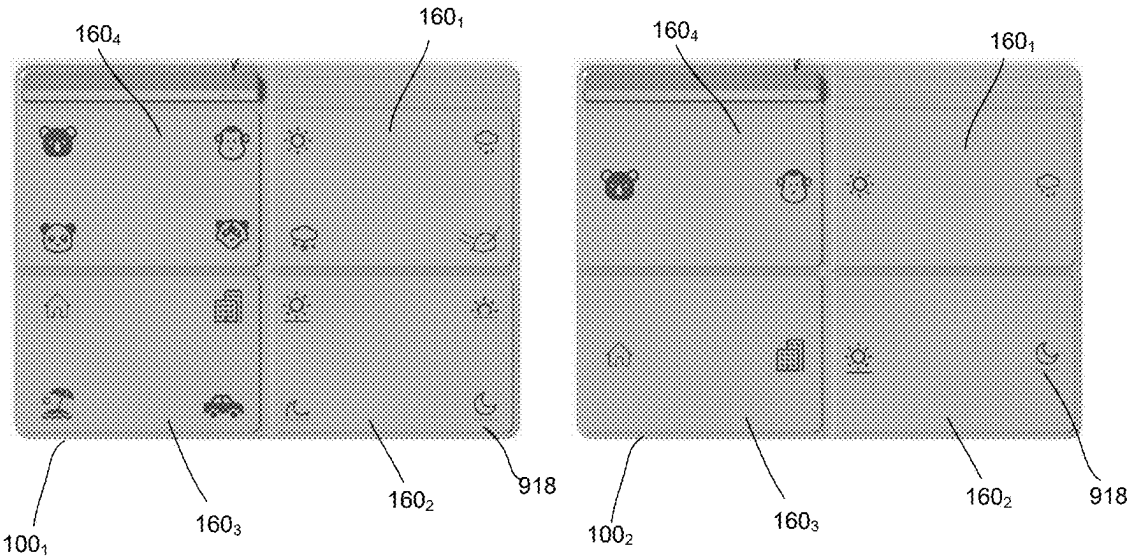


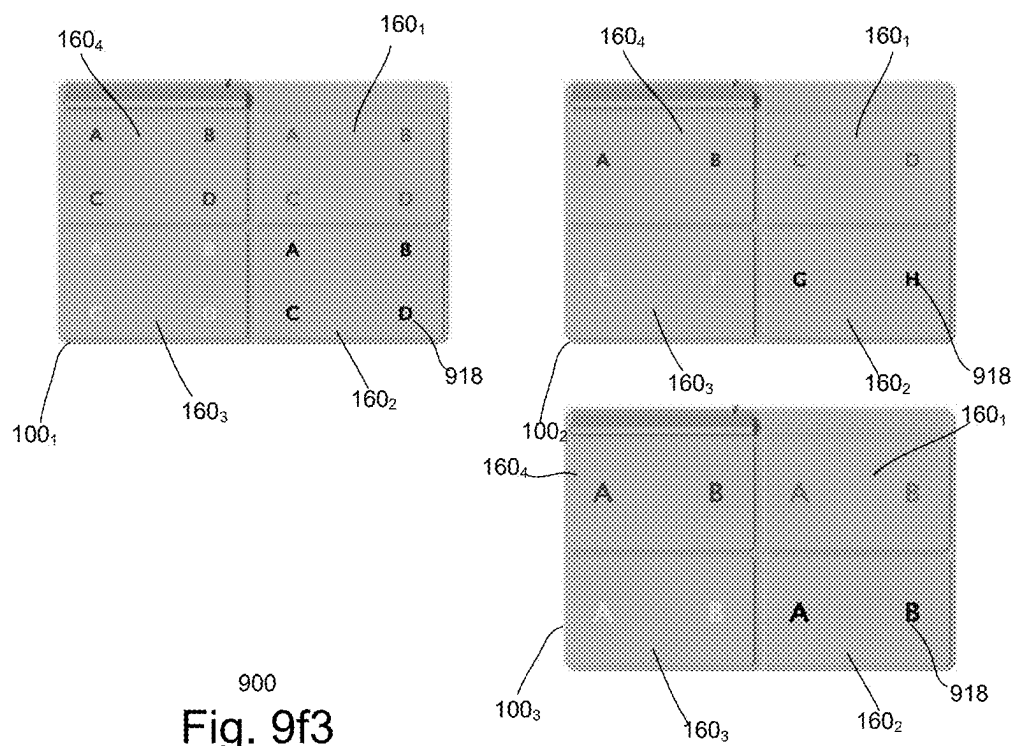
Fig. 9e2

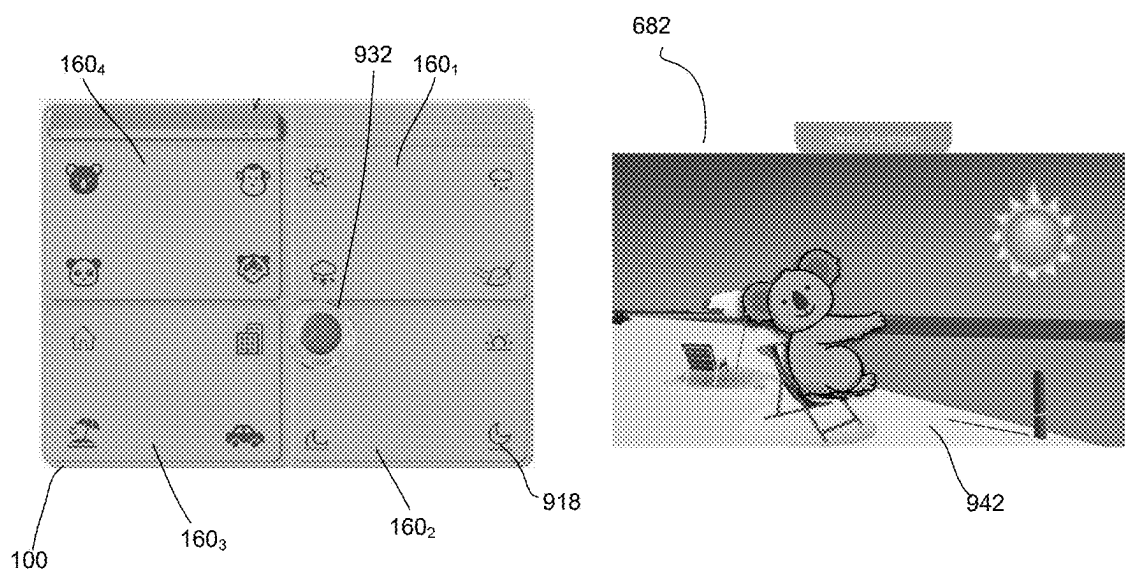


900
Fig. 9f1



900
Fig. 9f2





900
Fig. 9g

TOUCH SEQUENCE MAT REHABILITATION SYSTEM

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to U.S. Provisional Patent Application Ser. No. 63/562,270 filed on Mar. 7, 2024. This application also cross-references to co-pending U.S. patent application Ser. No. 18/242,538, filed on Sep. 6, 2023 which is a continuation of U.S. patent application Ser. No. 15,768,556 filed on Apr. 15, 2018. This application also cross-references to co-pending U.S. patent application Ser. No. 16/851,080, filed on Apr. 16, 2020 which is a continuation of U.S. patent application Ser. No. 14/118,772 filed on Feb. 17, 2014. All disclosures are herein incorporated by reference for all purposes.

FIELD OF THE INVENTION

[0002] The present disclosure generally relates to systems and methods for rehabilitating a patient through neuroplasticity.

BACKGROUND

[0003] Treating or rehabilitating a patient with a functional impairment, such as that resulting from a stroke, or a brain injury as well as psychological, mental, cognitive or physical disorders or disabilities, development delay, acute stress disorders, mild cognitive impairments (MCI) and similar types of disorders or impairments, traditionally involves repetitively performing tasks related to the functional impairment.

[0004] We have discovered that the mind, body and emotions have a positive effect on movement and neuroplasticity. For example, when body, mind and emotional state measures are at their respective target states, brain-body synergies occur, significantly enhancing rehabilitation through neuroplasticity. Body state, mind state and emotional state alignment resulting in brain-muscle synergies are described in, for example, U.S. patent application Ser. No. 18/242,538 and U.S. patent application Ser. No. 16/851,080, which are already herein incorporated by reference for all purposes.

[0005] The present disclosure is related to a portable sense mat configured to sense the pressure of motion of a patient when performing a physical activity to establish a compensation map. In particular, one or more portable pressure sensing pads or mats are used to generate a compensation map of a patient when the patient is performing an activity. The compensation map is analyzed spatially, temporally and synergistically across its various mat quadrants and auxiliary mat components to diagnose any weaknesses and loss of coordination in the movement of the activity. A personalized set of exercises is then prescribed to the patient to strengthen any identified weaknesses and re-establish coordination of the patient's movements in performing the activity. For example, the set of exercises can be used to facilitate rehabilitating the patient by repetitive practice.

SUMMARY

[0006] Effective treatment for accelerated recovery of patients with neurological damage or dysfunction is disclosed. In one embodiment, the present disclosure is related to a touch sequence pressure mat for rehabilitating a patient

with mobility, functional, coordination and cognitive issues due to neurological damage or disorder. The touch sequence pressure mat includes a flexible mat having top and bottom mat surfaces. The mat includes a functional region configured with pressure sensors for sensing pressure when the patient is performing prescribed exercises on the mat. The touch sequence pressure mat also includes a control module disposed on a controller region of the top mat surface. The control module is configured to provide power to the pressure sensors and to receive pressure information from the pressure sensors.

[0007] In another embodiment, the present disclosure is related to a method of rehabilitating a patient with mobility, functional, coordination and cognitive issues due to neurological damage or disorder. The method includes providing a touch sequence mat. The touch sequence mat includes a flexible mat having top and bottom mat surfaces. The mat includes a functional region configured with pressure sensors for sensing pressure when the patient is performing prescribed exercises on the mat. The touch sequence mat also includes a control module disposed on a controller region of the top mat surface. The control module is configured to provide power to the pressure sensors and to receive pressure information from the pressure sensors. The method also includes performing an activity of daily living (ADL) on the touch sequence mat by the patient. The method also includes generating a compensation map with spatial, temporal and magnitude pressure information of the patient performing the ADL. The method also includes analyzing the pressure information to identify compensation by the patient when performing the ADL. The method further includes prescribing a set of prescribed exercises for the patient to perform to correct for the compensation identified by analyzing the pressure information.

[0008] In another embodiment, the present disclosure is related to a method for forming a touch sequence mat. The method includes providing a flexible base mat layer, a rigid support layer, a flexible pressure sensor layer and a flexible top mat layer. The flexible pressure sensor layer includes a pressure sensor connector portion with a pressure sensor connector and the top mat layer includes a top surface cut out opening in a controller region of the top surface layer. The method also includes bonding the rigid support layer onto the flexible base mat layer. The method also includes bonding the flexible pressure sensor layer onto the rigid support layer where the flexible base mat layer, the rigid support layer and the flexible pressure sensor layer form a pressure sensor assembly for providing connections to the pressure sensors. The method also includes bonding the flexible top mat layer onto the pressure sensor assembly to form a mat stack where the top surface cut out opening is configured to accommodate the pressure sensor connector portion with the pressure sensor connector. The method also includes pressing the mat stack where pressing the mat stack transfers a rigid support layer pattern to a top surface of the flexible top mat layer. The method further includes mounting a control housing with a control module printed circuit board with a printed circuit board connector onto the controller region of the top mat layer. The pressure sensor connector is connected to the printed circuit board connector.

[0009] These and other advantages and features of the embodiments herein disclosed will become apparent through reference to the following description and the accompanying drawings. Furthermore, it is to be understood

that the features of the various embodiments described herein are not mutually exclusive and can exist in various combinations and permutations.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] In the drawings, like reference characters generally refer to the same parts throughout the different views. Also, the drawings are not necessarily drawn to scale, with emphasis instead generally being placed upon illustrating the principles of the invention. In the following description, various embodiments of the present invention are described with reference to the following drawings, in which:

[0011] FIGS. 1a-1b show different simplified views of an embodiment of a touch sequence mat;

[0012] FIG. 1c shows a simplified exploded view of an embodiment of a touch sequence mat;

[0013] FIG. 1d illustrates the portability of an embodiment of a touch sequence mat;

[0014] FIGS. 2a-2c show various simplified views of an embodiment of a control module of a touch sequence mat;

[0015] FIG. 2d shows a simplified block diagram components of an embodiment of the control module;

[0016] FIGS. 3a-3c show various layers of an embodiment of a touch sequence mat;

[0017] FIG. 3d shows an embodiment of a control module housing;

[0018] FIGS. 4a-4b show an embodiment of a block set for use with the touch sequence mat;

[0019] FIG. 4c illustrates an application of the block set with the touch sequence mat;

[0020] FIG. 5 shows an embodiment of a process flow for forming a touch sequence mat; and

[0021] FIGS. 6a-c illustrate embodiments of applications of a touch sequence mat;

[0022] FIG. 7 illustrates another application of an embodiment of a touch sequence mat set;

[0023] FIG. 8 shows a process flow of employing the touch sequence mat or mat set; and

[0024] FIGS. 9a-9g show various game applications for the touch sequence mat.

DETAILED DESCRIPTION

[0025] Embodiments relate to treating patients with neurological damage or disorders. In particular, a touch sequence (TS) mat is employed to diagnose weaknesses when performing activities and to prescribe a set of functional exercises to improve rehabilitation of the diagnosed weaknesses.

[0026] FIGS. 1a-1c show various simplified views of an embodiment of a touch sequence (TS) mat 100. FIG. 1a shows a simplified perspective view, FIG. 1b shows a simplified top view and FIG. 1c shows a simplified exploded view. Referring to FIGS. 1a-1c, the TS mat 100 includes a semi-flexible mat 110. The mat includes a top mat surface 160₇ having at least one function region 162. In one embodiment, the mat includes 4 functional regions 162₁₋₄. Providing a TS mat with other numbers of function regions may also be useful. For example, the TS mat may include any number of regions. In the case where there is more than 1 functional region, the functional regions are separated by functional region streets 164 on the top mat surface. The streets, for example, are recessed between the functional regions.

[0027] The mat with 4 functional regions may have a surface area of about 594×440 mm. As for a functional region, it may be about 284×185 mm. Other dimensions for the mat and functional regions may also be useful.

[0028] A functional region includes an array of pressure sensors for sensing the pressure of activity performed by the patient. The sensors may be spaced apart by a predetermined distance. For example, sensors are spaced apart by a distance of about 1.4 mm from the center. Other distances between sensors may also be useful, depending on the application requirements. The distance between the sensors determines the granularity of the sensor map. The closer the distance between sensors, the finer the granularity (higher spatial sensitivity); the farther the distance between the sensors, the coarser the granularity (lesser spatial sensitivity).

[0029] In some embodiments, software may be employed to provide the mat with selectable spatial resolution. The highest spatial resolution achievable is based on the actual physical distance between the sensors. Lower resolutions can be achieved by selecting only some of the sensors for sensing. For example, the next lower resolution would be selecting every other sensor for sensing. The software may be configured to include 2 resolution options, one being the highest achievable resolution and another being a lower resolution. Other configurations for mat sensing resolutions may also be useful.

[0030] The pressure sensors of a region are capable of fine pressure sensing. Each pressure sensor of the array in a functional region is configured to sense the amount or magnitude of pressure applied thereto. For example, the magnitude of the pressure sensed corresponds to a voltage generated by the sensor. In addition, the location of each sensor in the functional region can be spatially mapped, enabling a pressure map of the functional region to be generated. For example, the pressure magnitude and spatial distribution of the pressure results in a distribution graph or heat map.

[0031] The mat may be configured to sample the sensors at a rate of about 500 times per second. Other sampling rates, such as less than 500, may also be useful. Since the pressure data is a time-series data, the heat map transitions over time. Touch Sequence (TS) refers to the sequencing of the pressure data over time of the heat map. In the case of multiple function regions, pressure maps of all the regions can be generated. Each sensor, for example, represents a dot or pixel on the TS map. As such, the pressure data is both spatial and temporal. The map can be continuously generated during a rehabilitation session to track the patient when performing the functional activity.

[0032] It is understood that functional quadrants or regions need not all be operational. For example, depending on the functional activity being performed, one or more functional quadrants or regions of the mat may be switched off. Controlling the quadrants of the mat may be achieved by the control software.

[0033] As shown, the top surface of the mat includes a top surface controller region 165. The controller region, for example, is disposed on a side of the mat. As shown, the controller region is disposed at a top edge or periphery of the mat adjacent to a functional region. The controller region may be separated from a functional region by a region street. Providing the controller region at other edges of the mat or locations may also be useful. Preferably, the controller

region is disposed at a location of the mat which facilitates functionality, compactness and portability.

[0034] Disposed in the top surface controller region is a control module **170**. The control module, for example, includes a control module printed circuit board (PCB) **175** with electronic circuitry and components configured for controlling the operation of the mat. The control module PCB is contained in a control module housing **173**. The housing may be a rigid housing, such as an acrylonitrile butadiene styrene (ABS) housing. The housing may be formed by injection molding. Other types of control module housings may also be useful.

[0035] As for the control module PCB, it may be a rigid PCB. The thickness of the PCB may be about 0.8 mm. Other thicknesses may also be useful. A power cord **171** feeds power to the control module. The power cord can be connected to a port, enabling the power cord to be connectable to and disconnectable from the control module. Providing a permanently fixed power cord may also be useful. The control module may include a power indicator **177** control module housing. The power indicator may be a set of light pipes, such as red and green light pipes, which indicate when power is connected and when the mat is on or off. In some embodiments, the control module may include an internal portable power source, such as a battery. This enables the TS mat to function without external power. The dimensions of the housing may be about 276×29×7.4 mm while the control module PCB may be about 250×25×0.8 mm. Other dimensions for the housing and PCB may also be useful.

[0036] In addition, the control module can be configured to connect to a processing device, such as a desktop computer or a mobile processing device, such as a laptop, a tablet, or a smartphone. Other types of processing devices may also be useful. Connecting the processing device may be via a port, such as a USB port. Other types of communication ports may also be useful. The control module may be configured to connect to the processing device wirelessly. For example, a wireless communicator, such as a WiFi and/or a Bluetooth communicator may be employed. Other types of wireless communicators or combinations of wireless communications may also be useful.

[0037] The processing device may be employed to control the operation of the TS mat. For example, the processing device runs a mat control software, enabling a user to program the operation of the mat. The user may control, for example, the capture rate, the functional regions to capture, the spatial resolution, the capture rate of the sensors for generating pressure maps, when and how data is read out as well as other functions.

[0038] The top surface of the mat may include a top surface patterned region **166**. As shown, the top surface patterned region is disposed on the top edge of the mat adjacent to the top surface controller region. For example, the two regions mirror each other. The top surface patterned region may include a tactile embossed pattern. The embossed pattern may be a linear pattern with parallel lines parallel to, for example, the top edge of the mat. Other types of patterns may also be useful. The patterned region serves to aesthetically balance the right and left sides of the mat.

[0039] The mat, in one embodiment, is composed of multiple layers. As shown, the mat includes a base layer **120**, a support layer **130**, a pressure sensor layer **140** and a top layer **150**. The various mat layers are bonded to form the

mat. Bonding the various layers may be achieved with adhesives and pressing the layers of the stack together using a jig. Other techniques for bonding the various layers to form the mat may also be useful. The materials of the layers are configured to provide a TS mat which is durable and portable. For example, the mat is configured to be rigid and flexible where needed to produce a durable and portable mat.

[0040] The base layer may be a thermoplastic polyurethane (TPU) layer. The base layer may be about 0.5 mm thick. The base layer may include a fabric backing. The fabric, for example, may be a synthetic fabric to provide texture. Other types of fabric or having no fabric may also be useful. In one embodiment, the fabric side is the bottom side of the base layer which serves as the bottom mat surface **160_B**. Other thicknesses or configurations, including materials, of the base layer may also be useful.

[0041] The support layer is disposed above the base layer. The support layer may be an acrylonitrile butadiene styrene (ABS) layer. The support layer may be about 2 mm thick. Other thicknesses or configurations, including materials, of the support layer may also be useful. The support layer is generally a rigid layer. The support layer may be configured to include multiple support sub-layers to impart flexibility between the functional regions. The number of support layers may correspond to the number of function regions of the mat. For example, in the case of 4 functional regions, the support layer includes 4 support sub-layers. The configuration of the support layer provides rigidity for durability and support while the flexibility between the function regions facilitates portability.

[0042] The support sub-layers are configured to fit within the base layer. A jig may be used to position the support sub-layers on the base layer. In one embodiment, an adhesive is used to bond the support sub-layers on the base layer. For example, an adhesive may be applied to the back of the support sub-layers. The adhesive, for example, may be a double-sided adhesive tape. Alternatively, the adhesive may be a resin-based adhesive. Other types of adhesives or techniques for bonding the support sub-layers onto the base layer may also be useful. The support and base layers form a lower mat assembly of the mat.

[0043] As for the pressure sensor layer, it is configured with the pressure sensors. In one embodiment, the pressure sensor layer is a flexible PCB with pressure sensors. The flexible PCB may be about 0.3 mm thick. The pressure sensor includes a pressure sensor output connector portion with a pressure sensor connector. The connector provides connections to the pressure sensors of the pressure sensor layer. The connector is configured to connect to the control module connector on the control module PCB. The pressure sensor layer is bonded to the support layer. In one embodiment, double-sided adhesive tape stripes are used to bond the pressure sensor layer to the support layer. Other types of adhesives may be used to bond the pressure sensor layer to the support layer.

[0044] The top layer is disposed over the pressure sensor pad assembly. The top layer, for example, is similar or the same size as the base layer. The top and bottom layers sandwich the support and pressure sensor layers. In one embodiment, the top layer may be a fabric-backed TPU layer having a thickness of about 0.5 mm. The fabric, for example, may be a synthetic fabric to provide texture. In one embodiment, the fabric side is facing upwards and serves as the top surface while the TPU side faces the pressure sensor

assembly. The fabric provides a nice feel for the patient when performing the functional activity. Other configurations of the top layer, including material and dimensions, may also be useful.

[0045] In one embodiment, the top layer includes a cut out (opening) **169** located at the controller region **165**. The cut out allows the pressure sensor connector portion of the pressure sensor layer to be exposed, enabling the pressure sensor connector to be connected to the control module PCB. The top layer, in one embodiment, is shaped or patterned to produce the desired top layer pattern, such as the elevated functional regions **162**, the controller region **165**, the patterned region **166** and the functional region streets **164** therebetween. In one embodiment, the top layer pattern is produced by pressing the mat stack of the various layers of the mat. Other techniques for shaping the top layer may also be useful.

[0046] The control module **170** includes various components which are mounted onto the controller region **165** of the top layer of the mat. As shown, the control module includes a rigid control module PCB **175** with circuit components. The control module PCB, for example, may be about 0.8 mm thick. As for the planar dimensions, it may be about 250×25 mm. Other dimensions may also be useful.

[0047] A control module housing **173** encases the control module PCB. The housing, in one embodiment, is a rigid housing configured to protect the control module PCB. The control module housing, for example, may be a rigid housing or an acrylonitrile butadiene styrene (ABS) housing formed by injection molding. The dimensions of the housing, for example, may be about 276×29×7.4 mm. Other dimensions or configurations of the housing may also be useful.

[0048] As discussed, the mat layers form a mat of about 3.3 mm thick. This excludes the height of the control module housing. In one embodiment, the support layer provides rigidity of the mat while the separation of the sub-support layers impart flexibility for the mat.

[0049] FIG. 1*d* illustrates the portability of the TS mat **100**. As shown, the size of the TS mat can be reduced by folding the mat. For example, the original size of the mat may be about 594×440 mm. As shown, a first fold of the mat, along the width (longer direction) as illustrated by arrow A, the folded mat **106** results in a mat footprint **106** of about 243×594 mm. A second fold, as indicated by arrow B, then results in a folded mat **108** with a footprint of about 243×296 mm. As illustrated, the foldability of the mat results in a much smaller footprint, making it very portable.

[0050] FIGS. 2*a-2c* show various simplified views of the control module **170** on the mat **100**. FIG. 2*a* shows a simplified perspective view, FIG. 2*b* shows a simplified cross-sectional view along A-A' of the mat, and FIG. 2*c* shows a simplified view of the control module housing **173** with the control module PCB **175**.

[0051] Referring to FIG. 2*a-2c*, the control module **170** is fixed to a controller region **165** on a top surface **160_T** of the mat. For example, the control module includes a control module PCB **175** encased by a control module housing **173**. The mat is composed of multiple layers, such as the base layer **120**, the support layer **130**, the pressure sensor layer **140** and the top layer **150**.

[0052] The housing may be an injection molded ABS housing to encase the control module PCB. The dimensions of the housing, for example, may be about 276×29×7.4 mm.

Other dimensions or configurations of the housing may also be useful. As for the PCB, it is a rigid PCB with dimensions of 250×25×0.8 mm. The control module PCB is configured to fit within the control module housing. The control module housing is mounted onto the controller region by fasteners **272**, such as screws. As shown, the screws are fitted through the back surface **160_B** of the mat to mount the control module housing with the control module PCB.

[0053] As shown, the top layer includes a cut out which allows a pressure sensor connector portion **246** with a pressure sensor connector **247** to connect to the control module connector of the control module PCB.

[0054] As shown, a control module fastener **272**, such as a screw, attaches the control module housing with the control module PCB onto the top surface of the mat at the controller region. The fastener may mount the control module housing and PCB from the bottom surface **160_B** of the mat. In one embodiment, a gap exists between the bottom PCB surface and mat assembly and the top PCB surface and the control module housing. The gap, for example, may be at least about 0.2 mm. Other gap sizes may also be useful. For example, the gap between the bottom PCB surface and mat assembly may be about 1 mm while the gap between the top PCB surface and housing is about 4 mm.

[0055] A power cord **171** feeds power to the control module. The control module may include a power indicator **177**, indicating when power is connected and when the mat is on or off. In addition, the control module can be configured to connect to a processing device for programming and operating the TS mat.

[0056] FIG. 2*d* shows a simplified block diagram of the various units of the control module PCB **175**. The control module, as shown, includes a power unit **281**, a communication unit **282**, a control unit **284** and a pressure sensor unit **286**. In some embodiments, the control module PCB may also include a memory unit **288**. Providing the control module PCB with other units may also be useful. In some embodiments, the control module is configured to communicate with other TS-type mats. For example, the control module of the mat serves as a primary mat which communicates with secondary mats. Communicating with secondary mats can be via wired or wireless connections.

[0057] The power unit is coupled to, for example, the power cord **171**. The power cord can be connected to an external power source. The power cord may be connected to the power unit (control module) by a permanent connection. Alternatively, the power cord may be connectable to or disconnectable from the control module. For example, the port may be a power port. Other types of ports, such as USB ports may also be useful. The external power source may be an AC or DC power source. In the case of an AC power source, the power unit includes an AC-DC converter to convert the AC power to usable DC power for operating the mat. In the case of an external DC power source, there is no need to provide an AC-DC converter.

[0058] When power is connected to the power unit, the red indicator may light up, indicating that the mat is connected to power. When the on/off switch is switched on, the green light lights up, indicating that the mat is turned on. In some embodiments, the mat may be operational once the external power source is connected. In operational mode, power is provided to the different control module units, such as the communication, control, pressure and memory units.

[0059] The control unit controls the various units when the mat is connected to power and is switched on. For example, when the mat is switched on, power is provided to the various units to operate the mat. The control unit controls may be configured to control the operations of the communication, pressure and memory units.

[0060] The communication module may be configured to facilitate communication between the units of the control module with external devices connected to the control module. Communication with an external device may be achieved using wired or wireless sub-units. For wired sub-units, they may include USB ports or other ports which an external device can connect. For wireless sub-units, they may include Wi-Fi and or Bluetooth connections. Other types of connections with the external device and the control module may also be useful.

[0061] The control unit controls the various units of the control module. For example, based on instructions from the external device, the control unit controls the pressure sensor unit to detect pressure from the sensors to generate a pressure map. In addition, the control unit controls where the data is communicated. For example, the data may be stored in the memory unit for subsequent extraction or communicated to the external device via the communication unit.

[0062] In some embodiments, the control unit may include pre-settings for operating the mat. For example, settings could be selected, such as pressing the desired button setting, when no external devices are connected to control the operation of the mat.

[0063] In yet some other embodiments, the mat may be integrated with an external device. The external device may be used to analyze the pressure data collected. The data can be exported in real-time for analysis or for use by the external device. The data can be exported by WiFi or by Bluetooth via the communication unit of the control module. As also discussed, the data may be stored in the memory unit for subsequent exportation.

[0064] FIGS. 3a-c show simplified top views of various layers of a mat with four functional regions, such as that described in FIGS. 1a-1d. Common elements may not be described or described in detail.

[0065] Referring to FIG. 3a, a simplified top view of an embodiment of a lower mat assembly 301 is shown. The lower mat assembly includes a support layer 130 attached to a base layer 120. The base layer may be a flexible thermoplastic polyurethane (TPU) layer with a fabric backing. The dimensions of the base layer may be about 594×440×0.5 mm. Other dimensions or materials of the flexible base layer may also be useful. The fabric side of the base layer may be facing upwards while the non-fabric side of the base layer serves as the bottom mat surface.

[0066] The support layer is bonded to the top surface of the base layer. The support layer is a rigid support layer, such as an acrylonitrile butadiene styrene (ABS) layer. The thickness of the support layer may be about 2 mm thick. Other thicknesses may also be useful. To impart flexibility in the mat, the support layer is divided into separate or distinct support sub-layers, which together form the support layer. In one embodiment, the support layer is divided into 4 sub-layers 334₁₋₄, which correspond to the four function regions of the mat. This facilitates the resulting flexible mat which can be folded twice, as shown in FIG. 1d, to reduce the mat footprint to almost ¼ of the original or unfolded footprint.

[0067] As shown, the support sub-layers include two types of support sub-layers. Type A sub-layers 334₁₋₂ are bonded to the top half of the base layer and type B sub-layers 334₃ are bonded to the lower half of the base layer. A type A sub-layer includes a main portion defining a functional region of the mat. The main portion is a rectangular portion which is about the same size as a functional region of the mat. For example, the main portion is about 294×185 mm. Other dimensions may also be useful. A connector step portion 367 connects the main portion to a groove portion 365. The connector portion, for example, is configured to accommodate a connector portion of the pressure sensor layer with the pressure sensor connector. As for the groove portion, it corresponds to the controller or pattern region of the mat. As for a type B sub-layer, it includes only a main portion defining a functional region of the mat.

[0068] Referring to FIG. 3b, a simplified top view of an embodiment of a pressure sensor layer 140 is shown. The pressure sensor includes 4 functional pressure sensor regions, each with an array of pressure sensors 349. The dimensions of a functional region, for example, may be about 294×185 mm. Other dimensions may also be useful. A region may include about 693 sensors per region, with a total of 2772 sensors per pressure sensor layer. The sensors may be spaced apart by about 1.4 mm from the center. Other sensor spacings may also be useful.

[0069] The regions are interconnected by an inter-region connector portion 343 and an output connector portion 246 connected to a pressure sensor connector 247. The inter-region connector portion and the output connection portion facilitate connections to the pressure sensors by the pressure sensor connector.

[0070] The pressure sensor layer is configured to fit onto the support layer. In one embodiment, the functional pressure sensor regions are configured to fit onto the main portions of the type A and type B support sub-layers. As for the output connector portion, it is configured to fit onto the connector step portion of one of the type A support sub-layers.

[0071] The pressure sensor layer is configured to fit on top of the support layer. For example, the pressure sensor layer and support layer fit within the base layer. A jig may be used to position the pressure sensor layer on the support layer. In one embodiment, an adhesive is used to bond the pressure sensor layer to the support layer. For example, the adhesive is applied on the back surface of the pressure sensor layer. Other configurations for bonding the pressure sensor layer to the support layer may also be useful.

[0072] In one embodiment, the pressure sensor layer includes a pressure sensor connection portion with a pressure sensor connector. The pressure sensor connector provides connections to the pressure sensors of the pressure sensor layer. The pressure sensor connector is configured to connect to a control module connector on the control module PCB. The pressure sensor connection portion may not be glued to the connector step of the support sub-layer. The base layer, support layer and sensor layer form a pressure sensor assembly of the mat.

[0073] FIG. 3c shows a simplified top view of an embodiment of a top layer 150. The top layer may be the same size as the mat, including the size of the base layer. For example, the size of the top layer may be about 594×44 mm. The top layer forms a top surface 160_T of the mat. In one embodiment, the top layer may be a fabric-backed TPU layer having

a thickness of about 0.5 mm. Other thicknesses or configurations, including materials, of the top layer may also be useful. In one embodiment, the fabric side is facing the pressure sensor assembly. For example, the fabric side is the opposing side of the top surface of the mat.

[0074] In one embodiment, the top layer includes a cut out (opening) **169** located at the controller region **165**. The cut out allows the pressure sensor output connector portion of the pressure sensor layer to be exposed, enabling the pressure sensor output connector to be connected to the control module PCB. The top layer, in one embodiment, is shaped or patterned to produce the desired top layer pattern, such as the elevated functional regions **162**, the controller region **165**, a top layer connector step **368**, the patterned region **166** and the functional region streets **164** therebetween. In one embodiment, the top layer pattern is produced by pressing the mat stack of the various layers of the mat. Other techniques for shaping the top layer may also be useful.

[0075] FIG. **3d** shows a perspective view of a control module housing **173**. The control module housing is configured to fit on the controller region on the top layer. The control module housing encases the control module PCB. The dimensions of the housing may be about 276×29×7.4 mm while the control module PCB may be about 250×25×0.8 mm. Other dimensions for the housing and PCB may also be useful. The housing may be a rigid ABS housing. As shown, the control module housing may include a power indicator **177**. For example, the power indicator may be a set of light pipes, such as red and green light pipes, which indicate when power is connected and when the mat is on or off. An edge of the housing facing the functional regions includes a step opening **379** to accommodate the top layer connector step. The housing may also be configured to include communication ports for communicating with external devices as well as accommodating a power cord.

[0076] FIGS. **4a-4b** show an embodiment of a set of interlocking functional blocks **400** for use with the TS mat. Referring to FIGS. **4a-4b**, as shown, the set includes 3 blocks **410**, **420** and **430**. The blocks are configured to be able to interlock with any other block of the set. As shown, the first block **410** is a rectangular-shaped block while the second and third blocks, **420** and **430** are elongated blocks. In one embodiment, all the blocks have the same thickness **T**. The thickness **T**, for example, may be about 4.5 cm. Other thicknesses may also be useful. The blocks are preferably formed of a lightweight material to facilitate the portability of the system. The blocks, for example, may be elastomeric polymer (EVA) foam blocks. Other types of materials for the blocks may also be useful.

[0077] In one embodiment, the blocks of the set may be configured to be used alone as a single block or as a block assembly of any two blocks of the set when a patient is performing an activity. The blocks may serve to increase difficulty in performing the activity. In one embodiment, the blocks may be formed by having a soothing color. The soothing color, for example, may be turquoise. Other soothing colors or a combination of soothing colors may also be useful. By providing a soothing color for the blocks, it may serve to impart less sensory overload and promote calmness to the patient. The may facilitate the effectiveness of the rehabilitation. For example, putting the patient's mind state, emotional state and body state in to produce a synergistic effect.

[0078] Each block is configured with at least one interlocking recess **478**. In one embodiment, the first and second elongated blocks each include one interlocking recess disposed at about the center of the elongated blocks. The interlocking recesses, as shown, are rectangular-shaped recesses. As for the third block, it includes first and second interlocking recesses at about the center of adjacent sides. Other configurations of the blocks and interlocking recesses may also be useful. The interlocking recess of a block is configured to fit or interlock into the interlocking recess of another block to form a block assembly, as shown in FIG. **4b**. For example, the interlocking recesses facilitate forming a block assembly.

[0079] FIG. **4c** illustrates an example of using the block set on the TS mat **100**. As shown a block assembly **450** is disposed at about the center of the TS mat. The block assembly includes the first block **410** interlocked with the third block **430**. The first block, for example, is interlocked at a higher height, making it more difficult to perform the activity, such as moving the arm from one side of the mat to the other side. Other types of activities may also be performed. The first block can be interlocked at the lower height by rotating it to reduce the difficulty. In some cases, a single block can be used. This further reduces the difficulty level. The lowest difficult level would be, for example, not using any of the blocks. For example, a patient may drag an object on the mat in the absence of the block. However, the dragging will be captured by the mat due to the pressure signature.

[0080] In FIG. **5**, a simplified process flow **500** for assembling the TS mat is shown. The process assembles the various layers of the mat and control module, as described in FIGS. **1a-1d**, **2a-c** and **3a-3c**. Common elements may not be described or described in detail.

[0081] At **510**, the assembly process commences by providing a base layer. The base layer serves as the bottom layer of the flexible composite TS mat. For example, the composite mat layer includes multiple mat layers sandwiched together. In one embodiment, the base layer is a TPU layer with a fabric backing. The fabric side may be the bottom and the TPU side may be the top. The thickness of the layer may be about 0.5 mm. In one embodiment, for a mat with 4 functional regions, the mat may have a surface area of about 594×440 mm. For example, the dimensions of the base mat are 594×440×0.5 mm. Other configurations of the flexible base layer, such as material and size, may also be useful.

[0082] The assembly process continues to attach the support panel or layer to the base layer at **520**. To serve as a support layer, it is rigid. For example, the support layer is a rigid layer, such as an ABS layer with a thickness of about 2 mm. Other materials or thicknesses for the rigid support layer may also be useful. In one embodiment, to impart mat flexibility, the support layer is divided into distinct support sub-layers. For a mat with four functional regions, the support layer is divided into 4 sub-layers, such as that described in FIG. **3a**. For example, providing 4 support sub-layers enables the TS mat to be folded twice, as shown in FIG. **1d**, to reduce the mat footprint to almost ¼ of the original or unfolded footprint.

[0083] In one embodiment, the support sub-layers include 2 types, type A and type B. Type A sub-layers are disposed on a top half of the mat with the control module and patterned region while type B sub-layers are disposed on a bottom half of the mat. The sub-layers may be formed by

die-cutting a sheet of base layer material. Other techniques for forming the support sub-layers may also be useful.

[0084] The support sub-layers are configured to fit completely within the base layer. For example, an edge portion of the base layer surrounds the outer edge of the support sub-layers. In one embodiment, a jig is employed to position the sub-layers over the base layer. An adhesive is used to bond the support sub-layers to the base layer. The adhesive may be applied to the back of the support sub-layers. In some embodiments, the adhesive may be double-sided adhesive tapes. The double-sided adhesive tapes may be tape strips or patterned tape with the shape of the sub-layer. Other techniques for bonding the support sub-layers to the base layer may also be useful. The base and support layers form the lower mat assembly.

[0085] At **530**, the assembly process continues to attach the pressure sensor layer to the lower mat assembly. The pressure pads may include 4 functional regions with arrays of pressure sensors. The functional regions are interconnected by inter-region connector portions and an output connection portion is connected to a pressure sensor connector which is connected to a control module connector of the control module. The inter-region connector portion and the output connection portion facilitate connections to the pressure sensors by the pressure sensor connector.

[0086] The functional regions are configured to fit onto main portions of the support sub-layers. A jig may be used to align the pressure sensor layer onto the support sub-layers of the support layer. In one embodiment, double-sided adhesive tape is used to attach the pressure sensor layer to the support sub-layers. For example, the double-sided adhesive tape may be strips attaching the function regions of the pressure sensor layer to the main portions of the support sub-layers. Other types of adhesives may be used to attach the pressure sensor layer to the support layer. The pressure sensor layer and lower mat assembly form a pressure sensor mat assembly.

[0087] After attaching the pressure sensor layer onto the lower mat assembly, the assembly process continues to assemble a top layer to the pressure sensor mat assembly at **540**. The top layer may be about the same size as the mat, including the size of the base layer. For example, the top and base layers sandwich and encase the support and pressure sensor layers. The size of the top layer may be about 594×440 mm. The top layer forms a top surface of the mat. In one embodiment, the top layer may be similar to the base layer. For example, the top layer is a fabric-backed TPU layer having a thickness of about 0.5 mm. Other thicknesses or configurations, including materials, of the top layer may also be useful.

[0088] In one embodiment, the fabric side of the top layer faces the pressure sensor mat assembly while the TPU side serves as the top surface of the mat. In one embodiment, the TPU side may be a planar surface. In one embodiment, the top sheet includes a patterned region adjacent to the controller region. The pattern region is formed by hot stamping to emboss a tactile pattern, such as liner lines. As for the controller region of the top layer, it includes an opening to accommodate the output pressure sensor connector portion with the pressure sensor connectors.

[0089] The top layer is placed onto the jig to position it with the pressure sensor mat assembly. An adhesive may be applied to, for example, a back surface of the top layer and positioned by the jig. The jig, for example, ensures the

correct registration of the different layers. As discussed, the pressure sensor connector portion and pressure sensor connector are positioned through the opening in the controller region. The stack of layers with the jig are pressed together. In one embodiment, cold pressing is employed to press the assembly together. The pressure from the pressing causes the top layer to take the shape of the support sub-panels. For example, the top layer forms raised function regions separated by functional region streets.

[0090] The process continues for final assembly at **550** by attaching the control module PCB and control module housing to the mat. For example, the pressure sensor connector is connected to the control module connector, the control module PCG is positioned in the control module housing and the control module housing is mounted with the control module PCB using fasteners from the bottom mat surface. This completes the assembly process of the TS mat.

[0091] FIG. 6a shows an application **600** of the TS mat **100**. As shown, the TS mat is placed on the top surface of a desk or table **608**. The table may include a processing system with a monitor **682**. The processing system may be any type of computing system or mobile processing device, such as a laptop, tablet or smartphone. The patient may sit in a chair during a rehabilitation or treatment session to perform prescribed functional activities. The prescribed functional activities may be based on a compensation map established during a baseline or test session which identifies weaknesses in the patient's movements when performing certain activities. The prescribed functional activities are configured to rehabilitate the patient's identified weaknesses.

[0092] In one embodiment, functional activities, for example, may be related to activities of daily living (ADLs). The functional activities or exercises can be designed to enhance fine motor skills as well as gross motor skills. Fine motor skills, for example, involve the use of fingers to improve finger dexterity. Such activities may include grasping, pinching and controlled pressure applications. Gross motor skills, on the other hand, involve hand and arm movements.

[0093] As shown, the patient may sit on a chair to perform the prescribed functional activities on the TS mat on the table. The computer may be configured to display a set of functional activities for the patient to perform during the session. For example, the patient will imitate the motions displayed. For example, a rehabilitation program may be configured to display the prescribed functional activities for the patient to perform.

[0094] In some embodiments, the processing system is configured with sensors for monitoring either the mind state and body state measures or mind state, body state and emotional state measures of the patient while performing the prescribed functional activities. Should any of the states of the patient deviate from their target state measures, the system switches to an appropriate set of activities to put the deviated state measures back to their respective target state measures to produce synergistic effects from performing the prescribed functional activities, as described in U.S. patent application Ser. No. 18/242,538 and U.S. patent application Ser. No. 16/851,080, which are already herein incorporated by reference for all purposes.

[0095] FIGS. 6b-6c show other applications **600** of the TS mat using a desk or table **608**. Referring to FIG. 6b, the TS mat **100** is placed on a top surface of the desk **608**. A patient

602 is sitting on a chair 609 and is using the mat. As shown, the patient is fitted with sensors 680 and 684, as described in U.S. patent application Ser. No. 18/242,538 and U.S. patent application Ser. No. 16/851,080, which are already herein incorporated by reference for all purposes. For example, the sensors are configured to sense body and mind state signals. The patient may perform the exercises according to what is displayed on monitor 682. As for FIG. 6c, the patient with sensors 680 and 684, instead of sitting down, is lying on a bed 609 which has been inclined and performing exercises on the TS mat 100 placed on a table 608 according to, for example, what is displayed on monitor 682.

[0096] FIG. 7 shows another application 700 of an embodiment of a TS mat set. As shown, the TS mat set includes a plurality of mats configured for placement on various parts of a chair 709. For example, the mat set is a chair mat set. In one embodiment, the chair mat set includes a seat mat 712, a back mat 714, arm mats 716 and a foot mat 710. Providing other mats for the mat set may also be useful. The seat mat is configured to be positioned on the seat of the chair and the back mat is configured to be positioned on the back of the chair. The back mat may include a hanger to hang over the back of the seat. Other techniques for holding the back mat on the back of the chair may also be useful. For example, straps, such as Velcro-typed straps, may be employed. As for the arm mats, they are configured to be positioned on the arms of the chair. Straps may be employed to hold the arm mats in position on the arms of the chair. Other techniques for holding the arm mats in position may also be useful. The foot mat may be a single-foot mat to accommodate both feet or individual foot mats.

[0097] The mats of the set include pressure sensors, similar to the TS mat as described. Each mat of the set may be configured with a single functional region. Providing the mats with other numbers of functional regions may also be useful. The mats of the set may include control modules for providing power and to control the operation of the mats as well as to communicate with an external processing system. For the arm mats, due to their size, they may be provided with smaller-sized control modules. The mats may be constructed and assembled similarly to the TS mat. For example, the mats may have a base layer and a top layer sandwiching support and pressure sensor layers.

[0098] In one embodiment, the chair mat set is configured to operate with the TS mat as described. For example, the mats, as described in FIGS. 6-7, may be configured to operate together. In one embodiment, the TS mat on the table of FIG. 6 serves as a main or a primary mat while the mats of the chair mat set of FIG. 7 serve as auxiliary or secondary mats. In one configuration, the main mat may be connected to the external processing system while the secondary mats are wirelessly connected to the main mat, such as Bluetooth. Power to the secondary mats may be provided by portable power, such as by batteries in the control module. This reduces the number of wires needed. Other configurations of the primary and secondary mats may also be useful.

[0099] When the mats are used in conjunction, they provide a greater level of understanding of compensation by the patient. For example, even when movements involve only the arm, the patient may compensate by using the elbow or stronger muscles, such as the trunk or torso. Such compensation by the patient can easily be detected by the chair mats. In addition, the chair mats and TS table mat can be employed for activities, such as getting up from or sitting down on the

chair. The activity of getting up or sitting down may be accompanied by holding an object in one hand, such as a glass of water. In some cases, multiple floor or foot mats can be used for performing a walking activity.

[0100] FIG. 8 shows a simplified process flow 800 for diagnosing and treating a patient with, for example, neurological damage or disorder. Neurological damage and disorders may cause mobility, functional, coordination and/or cognitive issues in performing ADLs.

[0101] At 810, a compensation or TS map is generated. The TS map is for an ADL with which the patient has a mobility issue. For example, in the case of a stroke patient, the patient may have trouble with one side of the body, such as the right arm. The ADL may be related to the use of the right arm. In such cases, the compensation map may be generated using the TS table mat alone or with the TS chair mats. As discussed, when used in conjunction with the TS chair mats, a more comprehensive TS map can be generated. For example, the TS map can detect spatio-temporal feet, leg, trunk and arm weight shifts and weight bearing while trying to perform ADL. The TS table mat may serve as a primary TS mat and the TS chair mats may serve as secondary TS mats.

[0102] The TS table mat and TS chair mats may also be used for an ADL which involves getting up from or sitting down on a chair with an object in the hand. If the ADL involves walking, TS foot mats may be used. In general, in a typical sit-to-stand task, the impaired patient may first transfer excess weight onto the unaffected leg, tilt the body to the unaffected side and use one or both armrests for support. To get up, the patient may also push on the chair's backrest to build a forward momentum with the trunk in order to get up. By highlighting this sequence of compensation, the patient can progressively be trained to avoid using the backrest and armrests and to bear weight equally on both the seat and feet to stand without compensation. This can then be carried over into a stand-to-sit task. Similarly, for a typical reaching task with the arm, we may observe the patient using the backrest for support, tilting to the unaffected side and/or also using the opposite armrest for support. The patient can then be trained to disrupt this compensatory sequence and adopt a more functional and effective way of reaching for objects with the affected hand.

[0103] As an example, the ADL performed by the patient may be getting up from a chair with an object in his right hand, such as a glass of water. In performing the ADL, the patient loses balance and spills the water. The TS map generated from performing the ADL is analyzed at 820. The analysis of the TS map shows a compensated load was almost entirely on the left foot when the patient was getting up. The TS map shows that when getting up, there was an excessive pushback against the back of the chair, causing a jerky movement that resulted in the spilling of the water. The TS map also shows that while getting up, the patient put pressure on the left armrest, indicating that the left hand was used. Only after losing balance did the patient attempt to engage the right foot. This is an incorrect sequence. The right foot should be engaged first to help with balance, thus preventing the water from spilling.

[0104] After analyzing the TS map, the diagnosis indicates that the patient has a weak and unstable right foot. For example, the right foot should support the right hand and upper arm. Based on the diagnosis, a personalized set of exercises is prescribed to treat the patient at 830. The

personalized set of exercises should strengthen the right leg. This will improve weight bearing on the right leg as well as improve core strength so the chair backrest is not used unknowingly for trunk movement. In addition, this will correct sequencing and order of priority. For example, both right and left feet should be engaged first and then subsequently using the left armrest for balancing.

[0105] In some embodiments, the various exercises can be presented in audio-visual format on a mobile app or on a processing device to provide a comparison of a new map against the older map before the exercises start. In addition, an external device may be used to provide exercise treatment cycles, incorporating brain-muscle feedback. Furthermore, the portability and connectivity of the TS mat or mat system fulfills the repetitive practice requirements of neuroplasticity when a person has corrected his/her brain-muscle synergies and just needs complex movement repetitive practice for specific activities.

[0106] The TS mat or mat system can be employed to improve the rehabilitation of a patient with neurological damage or disorder causing mobility, functional, coordination and cognitive issues in performing ADLs. The TS mat system may be used for motor function rehabilitation using different treatment modalities, such as encouraging active hand use. The brain experiences learned non-use, such as avoiding using their weaker hand affected by the neurological disorder. The TS mat system counters that by providing visual and sensory feedback that encourages active participation by engaging their affected limb with targeted movements to prevent neglect. Both fine and gross motor skills can be improved by the TS mat system.

[0107] Hand-eye synchronization can be enhanced using the TS system. For example, a patient can follow visual cues on a screen to improve his/her ability to coordinate movements. This strengthens the neural connections between the brain, eyes, and hand movements, which are crucial for regaining daily functional abilities.

[0108] As already discussed, the TS mat system can reduce physical compensation. Many brain injury patients develop compensatory movement patterns where they use stronger muscles or the unaffected hand to complete tasks instead of rehabilitating the weaker side. The mat helps encourage isolated movement of the affected hand or limb by requiring precise pressure application. A patient's movement compensation can be identified using the TS map. Exercises can be prescribed to train proper movement mechanics and timing by prompting the patient to engage the different parts of the body in the correct sequence, speed and timing. For example, the diagnosis and treatment using the TS mat focuses on how the patient interacts with the external world. Long-term imbalances that can lead to secondary musculoskeletal issues, such as shoulder pain and joint stiffness, can be prevented.

[0109] The TS system can also improve trunk control. Good trunk stability is essential for controlled arm and hand movements. Many patients, especially stroke survivors, struggle with weak core muscles which leads to excessive leaning or imbalance. The mat supports trunk control by encouraging a stable seated posture when performing arm or hand movements. It requires symmetrical weight distribution, provides balance challenges (such as reaching across quadrants) to activate core muscles, and promotes better coordination between trunk and arm movement which is essential for everyday tasks like reaching for objects.

[0110] Body alignment and postural awareness can be enhanced using the TS mat system. Proper body alignment is crucial for effective rehabilitation and to avoid strain on unaffected body parts. The mat helps with guiding the patient to maintain an upright posture during exercises. This includes preventing excessive leaning or slouching, which can limit arm function. Also, the TS mat system encourages symmetrical movement patterns, which reduces dependency on the stronger side as well as reinforces good positioning habits that translate into ADLs.

[0111] The TS mat system can be programmed to provide adaptive and progressive training based on the progress of the patient. For example, exercises can start with simple hand placements and gradually progress to more complex motion sequences. The exercises can be adjusted for different levels of impairment, ensuring a customized rehabilitation approach. This helps train reaction time and movement precision, making rehabilitation more effective.

[0112] In addition, the TS mat system can be configured to support bilateral coordination. For example, the mat can be used to train both hands in coordinated movements, encouraging balanced development. This is particularly useful for stroke patients to regain symmetrical hand function. Also, the system can be used to help retrain grip strength and pressure control. For example, by mapping pressure points and movement patterns, a therapist can assess how much force a patient is applying. This helps in tracking sensory improvement over time. The exercises can be customized for different levels of impairment. Regaining grip control is essential for holding objects, writing, and performing other daily tasks.

[0113] In one embodiment, the TS mat system can be used for cognitive engagement and training. The TS mat system can be a powerful tool for cognitive rehabilitation, helping patients with a brain injury to improve their mental functions while engaging in physical exercises. By integrating cognitive tasks with movement, the TS mat system supports neuroplasticity, strengthening brain connections essential for recovery.

[0114] The TS mat system can enhance memory and recall abilities. The TS mat system can present touch sequences that a patient must remember and repeat, improving the working memory. Gradual increases in complexity can help strengthen short-term and long-term recall. This is useful for patients with traumatic brain injury (TBI) or stroke-related memory deficits.

[0115] The attention and focus of the patient can also be improved. For example, a patient follows visual and auditory cues while completing touch sequences. This helps to improve concentration and sustained attention, which are often impaired after brain injuries. In addition, the prescribed exercises can be customized with distraction-based challenges to train selective attention.

[0116] The TS mat system can be configured to develop problem-solving and decision-making skills. For example, some exercises can include pattern recognition or logic-based sequences, which help train critical thinking. A patient may have to decide which quadrant to press next based on a set of rules, enhancing decision-making abilities. This encourages adaptive thinking by introducing changing patterns and unexpected challenges.

[0117] Hand-eye coordination and spatial awareness can be boosted using the TS mat system. The TS mat system can be configured with activities for a patient to perform such as

targeting specific quadrants, thereby improving visual-spatial processing. Other activities, like navigating spaces, recognizing distances, and coordinating movements are also useful. This is beneficial for patients recovering from perceptual deficits due to brain injuries.

[0118] Multisensory Integration can also be stimulated by using the TS mat system. The TS mat system can incorporate tactile, visual, and auditory feedback, enhancing sensory processing. This may include exercises requiring matching colors, sounds, or touch points to help a patient integrate multiple senses simultaneously. This supports overall cognitive flexibility, a key aspect of rehabilitation.

[0119] In some embodiments, the TS mat is configured for gamified training for motivation and engagement. Cognitive exercises can be made interactive and game-like, increasing motivation. A patient receives scores, progress tracking, and positive reinforcement, encouraging continued participation. The use of adaptive difficulty levels ensures that the training remains both challenging and achievable.

[0120] FIGS. 9a-9g show various game applications for the TS mat. Referring to FIGS. 9a-9c, the TS mat 100 may be configured with removable or replaceable stickers 918 on each quadrant 160₁₋₄. For example, as shown in FIG. 9a, the stickers are letters on each quadrant. Target spots could be used for simple training instructions. For example, the instructions may be to move an object from quadrant A to quadrant B, or from quadrant C to quadrant B. Other instructions for moving the object from one quadrant to another may also be useful. The pressure sensors of a quadrant can detect if the task has been performed correctly before moving forward accordingly.

[0121] As shown in FIG. 9b, the stickers may be stickers of different shapes, such as a circle, square, triangle and a 3-pointed star. Other shapes may also be useful. Similar to FIG. 9a, instructions can be issued to move the object from one shaped sticker to another shaped sticker. For example, the object is moved from a quadrant with a circle sticker to a quadrant with a square sticker.

[0122] In another game, as shown in FIG. 9c, each quadrant includes four stickers of the same shape but different colors, with each quadrant featuring a distinct shape. For example, the first quadrant 106₁ is provided with four square-shaped stickers of different colors, the second quadrant 106₂ is provided with four 3-pointed shaped stickers of different colors, the third quadrant 106₃ is provided with four triangular-shaped stickers of different colors, and the fourth quadrant 106₄ is provided with four circular shaped stickers of different colors. Target spots can be used for more complex training instructions, such as moving an object from the black circle to the red circle.

[0123] FIGS. 9d1-9d2 show another game 900. The game may be called "Feed the Koala". Referring to FIG. 9d1, a TS mat 100 has palm tree stickers 918 on corners of the quadrants 160₁₋₄. An object 932, such as a bottle, is placed on the mat. Other types of objects may also be useful. A display 682 of a computer device shows a user interface that illustrates the four quadrants of the mat. An instruction is issued to the patient to move the bottle to the palm tree at the lower right corner of the fourth quadrant 160₄. Once the task has been successfully completed, a koala 942 appears on the palm tree at the lower right corner of the fourth quadrant of the mat on the user interface on the display 682 of the computer device.

[0124] In FIG. 9d2, an instruction is issued to the patient to move the bottle from the lower right corner of the fourth quadrant to the upper left corner of the fourth quadrant. After the task has been successfully completed, the koala 942 appears on the palm tree at the upper left corner of the fourth quadrant on the user interface of the display 682 of the computer device. This indicates that the patient has successfully completed the task.

[0125] FIGS. 9e1-9e2 illustrates another game 900 using the TS mat 100. Referring to FIG. 9e1, the quadrants 160₁₋₄ of the mat each include four stickers 918 of different shapes but the same color. The colors of the stickers of each quadrant are different. An object 932 is placed on the mat. A display 682 of a computer device shows a user interface illustrating various shaped objects of different colors corresponding to some of the objects on the mat.

[0126] In FIG. 9e2, an instruction is issued to the patient to move the object onto the yellow triangle (which is located in the lower-left corner of the third quadrant 160₃). The user interface on the display 682 will indicate 942 that the patient has successfully completed the task upon successful completion of the task.

[0127] FIGS. 9f1-9f3 show different configurations of the mat with stickers corresponding to different complexity levels. Referring to FIG. 9f1, first and second mats 100₁₋₂ are shown. The mats each include four quadrants 160₁₋₄. The mats include different-shaped stickers 918 of the same color in each quadrant. As shown, the stickers on the first mat are configured to be more complex than the stickers on the second mat. For example, the first mat includes four stickers for each quadrant which are of the same color but different shapes. In addition, the colors of each quadrant are different. On the other hand, the second mat only has two stickers for each quadrant which are of different shapes but the same color, with each quadrant having a different color.

[0128] FIG. 9f2 shows another configuration of first and second mats 100₁₋₂ with different complexity levels. The mats each include four quadrants 160₁₋₄. The mats include different-shaped stickers 918 for all the quadrants. In other words, each sticker on the mat is unique. As shown, the stickers on the first mat are configured to be more complex than the stickers on the second mat. For example, the first mat includes four stickers for each quadrant while the second mat includes only two stickers for each quadrant.

[0129] Referring to FIG. 9f3, another configuration of first, second and third mats 100₁₋₃ with different complexity levels is shown. The mats each include four quadrants 160₁₋₄. The mats include different letter stickers 918 for all the quadrants. As shown, the stickers on the first mat are configured to be more complex than the stickers on the second mat and the third mat is configured to be the least complex. For example, the first mat includes the same set of four stickers for each quadrant with each quadrant being of a different color. As for the second mat, each quadrant includes a unique set of two stickers, with each set being of a different color. Similar to the second mat, the third mat includes the same set of two stickers for each quadrant with each set being of a different color. However, the size of the stickers on the third mat are larger than the size of the stickers on the first and second mats.

[0130] FIG. 9g shows another game 900 using the TS mat 100. As shown, the TS mat is configured with 4 unique stickers 918 in each quadrant 160₁₋₄. A patient can move an object 932 to a desired sticker to create a fun scene 942

which will be displayed on the user interface on display 682. For example, if the patient moves the object to the sun sticker, the fun scene created will include the sun, such as a koala tanning on a beach on a sunny day.

[0131] As discussed, the TS mat system is equipped with real-time feedback and progress tracking. For example, movements are displayed on a screen for instant feedback. Data is recorded to monitor the patient's progress over time. This assists in adjusting the prescribed exercises over time based on performance.

[0132] The portability of the TS mat system makes it easy to store and transport. The TS mat system can be used anywhere, for example, in hospitals, therapy centers and even at home. The adaptability of the exercises makes the TS mat system suitable for different levels of recovery and age groups of the patient.

[0133] The present disclosure may be embodied in other specific forms without departing from the spirit or essential characteristics thereof. The foregoing embodiments, therefore, are to be considered in all respects illustrative rather than limiting the invention described herein. The scope of the invention is thus indicated by the appended claims, rather than by the foregoing description, and all changes that come within the meaning and range of equivalency of the claims are intended to be embraced therein.

1. A touch sequence pressure mat for rehabilitating a patient with mobility, function, coordination and cognitive issues due to neurological damage or disorder comprising:

a flexible mat having top and bottom mat surfaces, wherein the mat includes a functional region configured with pressure sensors for sensing pressure when the patient is performing prescribed exercises on the mat; and

a control module disposed on a controller region of the top mat surface, the control module is configured to provide power to the pressure sensors and to receive pressure information from the pressure sensors when the patient is performing the prescribed exercises on the mat.

2. The touch sequence pressure mat of claim 1 wherein the control module is configured to connect to an external processing system which controls the operations of the touch sequence pressure mat.

3. The touch sequence pressure mat of claim 2 wherein the control module is configured to transfer pressure information from the pressure sensors to the external processing system.

4. The touch sequence pressure mat of claim 2 wherein the prescribed exercises are displayed on a display of the external processing system for the patient to follow to perform the prescribed exercises.

5. The touch sequence pressure mat of claim 1 comprises a plurality of functional regions.

6. The touch sequence pressure mat of claim 1 comprises four functional regions arranged in a 2x2 matrix.

7. The touch sequence pressure mat of claim 6 wherein the four functional regions are individually controllable functional regions.

8. The touch sequence pressure mat of claim 1 wherein the pressure information comprises spatial, temporal and magnitude pressure information of the patient performing the prescribed exercise.

9. The touch sequence pressure mat of claim 6 wherein: the flexible mat with the four function regions comprises a foldable mat along functional region streets separating the four functional regions; and

the functional mat is capable of folding twice to reduce a footprint of the mat by almost about $\frac{1}{4}$.

10. A method of rehabilitating a patient with mobility, function, coordination and cognitive issues due to neurological damage or disorder comprising:

providing a touch sequence mat comprising

a flexible mat having top and bottom mat surfaces, wherein the mat includes a functional region configured with pressure sensors for sensing pressure when the patient is performing prescribed exercises on the mat, and

a control module disposed on a controller region of the top mat surface, the control module is configured to provide power to the pressure sensors and to receive pressure information from the pressure sensors;

performing an activity of daily living (ADL) on the touch sequence mat by the patient;

generating a compensation map with spatial, temporal and magnitude pressure information of the patient performing the ADL;

analyzing the pressure information to identify compensation by the patient when performing the ADL; and

prescribing a set of prescribed exercises for the patient to perform to correct for the compensation identified by analyzing the pressure information.

11. The method of claim 10 wherein:

performing the prescribed exercises are performed on the touch sequence mat; and

collecting spatial, temporal and magnitude pressure information of the patient performing the prescribed exercises.

12. The method of claim 10 wherein providing the touch sequence mat comprises:

providing a touch sequence mat set comprising touch sequence mats to be placed on a chair, including a backrest, a seat, armrests and at a foot of the chair; and performing the ADL which comprises the patient performing the prescribed exercises on the chair with the touch sequence mats.

13. The method of claim 10 wherein;

the touch sequence mat is disposed on a top surface of a table; and

performing the ADL which comprises the patient moving an affected arm.

14. The method of claim 10 wherein the touch sequence mat comprises four functional regions with pressure sensors separated by functional region streets, the four functional regions are arranged in a 2x2 matrix.

15. The method of claim 14 comprises folding the touch sequence mat twice along the functional region streets to reduce a footprint of the touch sequence mat to almost about $\frac{1}{4}$.

16. The method of claim 10 wherein the prescribed exercises comprise gaming exercises.

17. A method for forming a touch sequence mat comprising:

providing a flexible base mat layer, a rigid support layer, a flexible pressure sensor layer and a flexible top mat layer, wherein

the flexible pressure sensor layer includes a pressure sensor connector portion with a pressure sensor connector, and

the top mat layer includes a top surface cut out opening in a controller region of the top surface layer;

bonding the rigid support layer onto the flexible base mat layer;

bonding the flexible pressure sensor layer onto the rigid support layer, wherein the flexible base mat layer, the rigid support layer and the flexible pressure sensor layer form a pressure sensor assembly for providing connections to the pressure sensors;

bonding the flexible top mat layer onto the pressure sensor assembly to form a mat stack, wherein the top surface cut out opening is configured to accommodate the pressure sensor connector portion with the pressure sensor connector;

pressing the mat stack, wherein pressing the mat stack transfers a rigid support layer pattern to a top surface of the flexible top mat layer; and

mounting a control housing with a control module printed circuit board with a printed circuit board connector onto the controller region of the top mat layer, wherein the pressure sensor connector is connected to the printed circuit board connector.

18. The method of claim **17** wherein:

the rigid support layer comprises four rigid support sub-layers corresponding to four functional regions of the touch sequence mat with pressure sensors;

the rigid support sub-layers are arranged in a 2×2 matrix on the base layer to form the touch sequence mat with a 2×2 matrix of 4 functional regions; and

the touch sequence mat is configured to fold along functional region streets separating the four functional regions to reduce a footprint of the mat by almost about $\frac{1}{4}$.

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